# SE(3)-WAM: What Should a Robot Remember and Predict for Dynamic Manipulation? Supplementary Material

Anonymous CVPR submission

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This supplement documents the measured experiments, implementation, and remaining study protocols. Sections A–D give the data, configurations, and results; E–F specify pending comparisons; G presents diagnostics and offline predictions; H reports the completed recipe and frozen-decoder studies.

| Appendix | Contents | Main |
|---|---|---|
| A | Tasks, data, information, scoring | Sec. 3 |
| B | Implementation and system budgets | Secs. 4–5 |
| C | History, frozen readouts, interventions | Sec. 5.4 |
| D | Future targets, statistics, combinations | Secs. 5.2–5.3 |
| E | Backbone and information flow | Secs. 4.4, 5.6 |
| F | Pretraining, deployment, external tests | Sec. 5.6 |
| G | Failures and qualitative evidence | Sec. 5.5 |
| H | Training recipes and frozen decoding | Secs. 5.3, 5.5 |

## A. Tasks, Data, and Evaluation

### A.1. The Reported Fourteen-Task Suite

The completed studies use the migrated DB0 v0.6 demonstration pool and its fourteen-task evaluation suite. Table S1 lists the task identifiers used in the result arrays. The task-balanced score includes every task, including zero-success conditions.

The common training pool has 11,200 episodes, 800 per task. The internal representation screen uses 350 validation episodes; adapted systems retain their own validation settings (including 1,400 for DreamVLA/OpenWAM). The future screen samples 160,416 chunks per epoch. Stored episode count and effective chunk exposure measure different aspects of training. Later fresh-data production retains its own data identity and does not retroactively change the source of these results. Motion-only P0 prediction has no corresponding action-learning score in the fourteen-task table.

Success is read from the native task evaluator after its ordered physical requirements (Table S1). The accepted FUTURE0, BASE2, and HISTORY1 packages reference the same archived evaluator source, task configurations, and fourteen-suite scenario manifest. We recover the definitions from that archive. Success is checked at each 2 ms physics step. Interrupted continuous conditions reset their counters. For T2, the sample completing the low-relative-motion counter also starts the stable-dwell counter; the two durations should not be added as an exact 0.600 s requirement.

**Terminal events.** The native result mapping groups events into five categories: *invalid* (invalid state); *boundary* (workspace or downstream exit, drop, expired goal window); *safety* (unsafe contact, excessive rim impulse, wire contact, stage block, release impulse, early robot contact, tray or gripper–support contact, striker trap); *task violation* (blocked driver, release outside the goal, path deviation or backtracking, premature release, stopping/deflecting without capture, unstable tipping, missing/extra impact, wrong striker contact, or insufficient outgoing motion change); and *timeout*. These categories describe evaluator outcomes.

There is no global priority among categories. Each family selects the first terminal event in its physics-step evaluation order and preserves that result. Grasp checks invalid state, workspace exit, downstream exit, and drop before success, then timeout. T1 inserts unsafe occluder contact after invalid state. T2 checks invalid state, rim impulse $> 0.10\,\mathrm{N\,s}$, drop, goal-window expiry, and release outside the goal before success and timeout. T3 checks invalid state, stage block, drop, workspace exit, wire contact, premature release, path deviation, and backtracking before success and timeout. Wire failure uses impulse $> 0.020\,\mathrm{N\,s}$ or penetration $> 0.004\,\mathrm{m}$ for more than two consecutive steps; path deviation is $> 0.030\,\mathrm{m}$ for 0.050 s, and backtracking exceeds 0.05 relative to maximum progress.

T4 checks invalid state, release impulse $> 0.002\,\mathrm{N\,s}$, unsafe ramp contact, unsupported tray contact, can tipping, early-contact/boundary branches, and invalid supported capture before success and timeout. A target that has exited

Table S1. Native success conditions in the reported fourteen-task suite. Thresholds come from the evaluator source and task configurations bound to the accepted FUTURE0, BASE2, and HISTORY1 packages. A task must pass its preceding stages and avoid terminal failures. All tasks run at 20 Hz with a 2 ms physics step; durations below refer to physical time. COM denotes center of mass.

| Task | Ordered success conditions | Limit |
|---|---|---|
| grasp; t1_belt, t1_xyz, t1_so3, t1_occ | Approach within 0.10 m; bilateral fingerpad contact for 0.05 s; collision-geometry bottom at least 0.05 m above support; retain contact and clearance for 0.50 s. T1 adds conveyor drive, translation, rotation, or a physical occluder. SO3 has no additional success-angle threshold; unsafe occluder contact terminates OCC. | 160 steps 8 s |
| t2_trans | Approach within 0.10 m. Entire object collision geometry inside the receptacle: planar clearance and floor gap both $\geq -0.002$ m, top no higher than the opening. Release requires an opening command $< -0.5$ and no fingerpad contact. Relative linear/angular speeds $\leq 0.08$ m/s and $0.8$ rad/s for 0.10 s, followed by a 0.50 s stable dwell. | 200 10 s |
| t2_se3 | Approach within 0.05 m and 0.12 rad. Full T-shaped containment: planar clearance $\geq 0.002$ m, floor gap in $[-0.002, 0.002]$ m, tilt $\leq 0.05$ rad, top below opening. After release, relative speeds $\leq 0.02$ m/s and $0.04$ rad/s for 0.10 s, then a 0.50 s dwell. The 0.12 rad gate applies to approach. | 200 10 s |
| t3_phase | Start with the native pre-grasp. Phase alignment: distance $\leq 0.090$ m and loop-direction error $\leq 0.20$ rad. Enter within 0.030 m at progress $\leq 0.15$; reach historical maximum progress $\geq 0.60$, then $\geq 0.85$ and endpoint distance $\leq 0.055$ m. | 420 21 s |
| t3_full | Acquire the wand with bilateral contact for 0.05 s, then complete the phase stages above. In a later sample, retain current progress $\geq 0.85$ and move at least 0.045 m from both path and endpoint. Finally open ($< -0.5$) with no fingerpad contact. Releasing before exit fails. | 640 32 s |
| t4_sphere | Approach within 0.10 m; first qualifying contact after ramp exit; any tray contact before success fails. Bilateral hold for nominal 25 ms (12 physics samples, 24 ms). Raise COM by 0.05 m from its height when the bilateral-hold criterion is first met, and retain contact and height for 0.50 s. | 120 6 s |
| t4_sphere_tabletop; t4_slider | Approach within 0.10 m; contact after ramp exit, while supported, in the capture region, and moving at planar speed $\geq 0.08$ m/s. Bilateral hold for 0.05 s, COM lift $\geq 0.05$ m from the first attainment of that hold criterion, then contact and height for 0.50 s. | 120 6 s |
| t4_can | Same supported-capture sequence. Before first fingerpad contact, a can-axis absolute dot product with vertical $> 0.45$ fails as unstable tipping. The configuration does not require a transverse closing axis. | 120 6 s |
| t5_replan | Exactly one selected-side impact group completes before robot contact. At 30 ms after the physical impact-end timestamp, branch-signed lateral speed $\geq 0.035$ m/s and yaw rate $\geq 0.35$ rad/s. Approach within 0.10 m; bilateral hold with grasp-axis error $\leq 18^\circ$ for 0.05 s; COM lift $\geq 0.05$ m from the first attainment of that hold criterion and stable hold for 0.50 s. Impact-group lag is 12 ms and maximum contact is 0.10 s. | 120 6 s |

the ramp, remains in contact with the downstream support, and has no fingerpad contact at speed $\leq 0.03$ m/s for 0.25 s fails as stopping/deflecting only. T5 checks invalid state, unsafe contact, early robot contact, wrong striker, extra impact/striker trap, insufficient outgoing change, and boundary events before success. At the horizon, zero completed impacts yields missing selected impact; otherwise the result is timeout.

## A.2. Scenario Sampling and Replay

The historical E7/agentview recipes specify proposal ranges (Table S2). Continuous min–max fields use uniform draws from NumPy PCG64; discrete choices are equiprobable before physical-feasibility checks. Evaluation loads each accepted scene's complete stored ScenarioSpec and does not resample its parameters. For harmonic translation,

$$p(t) = c + (0, v_y, 0)t + (A_x \sin(2\pi f_x t + \phi), 0, 0),$$

where $\phi \sim U[-\pi, \pi]$. Thus $c$ differs from the initial position by the phase-dependent offset. T4 targets start at zero linear/angular velocity and evolve under free dynamics; no constant target speed is prescribed. Their slope is continuously sampled from $[5, 15]^\circ$. Ramp-local offsets use the ramp center as origin and lateral, downhill, and normal axes: $p_{\text{world}} = c_{\text{ramp}} + R_x(-\alpha)p_{\text{offset}}$. The center is derived from the exit lip and ramp half-size; offsets are not added directly to the lip. T2-SE3 training uses $\omega_z \sim U[-.25, .25]$; validation excludes $(-.08, .08)$. The other thirteen tasks share train/validation proposal recipes.

Occlusion acceptance requires at least eight initially visible frames with visible fraction $\geq .95$, ten to sixty fully

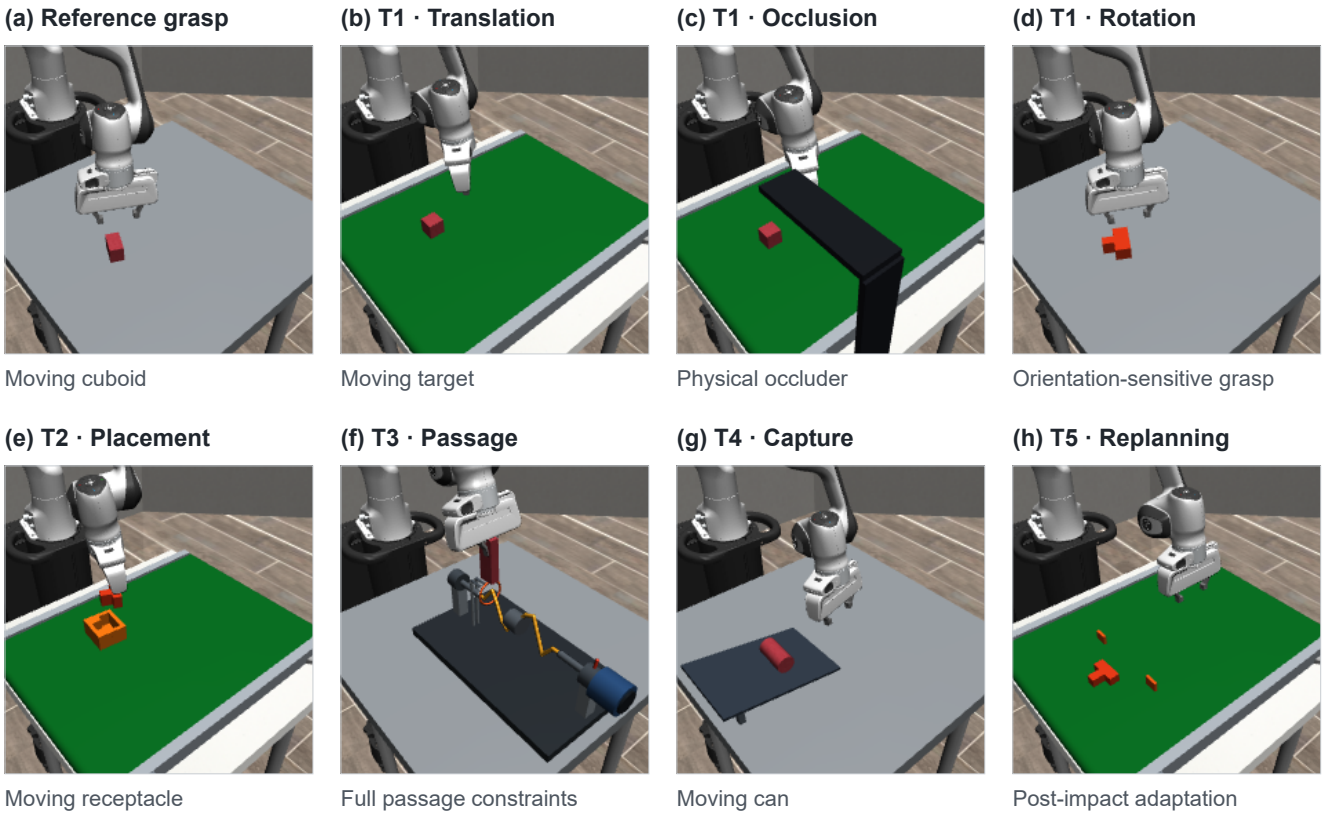



Figure S1. Recorded benchmark observations. Eight examples cover reference grasping; T1 translation, physical occlusion, and orientation-sensitive grasping; T2 moving-receptacle placement; T3 constrained passage; T4 capture; and T5 post-impact adaptation. For each displayed task, the lowest state index in the accepted DEV84 manifest supplies the final observed, uncropped $224 \times 224$ agentview RGB. Labels identify task conditions; the snapshots illustrate appearance and geometry, while task outcomes are reported separately. Table S1 lists all fourteen tasks.

occluded frames (fraction $\leq .05$, lasting 0.5–3.0 s), and at least three reappearance frames with fraction $\geq .8$. These are acceptance criteria, not independent sampling variables. For example, the accepted OCC scenes have belt speeds 0.059588–0.099852 m/s within the wider proposal interval. Accepted source seeds can have gaps; a consecutive seed range does not reconstruct the scene set.

All 1,400 stored scenarios share the robot joint initialization

$$q_0 = [0,\ .19634954,\ 0,\ -2.61799388,\ 0,\ 2.94159265,\ .78539816]\ \text{rad},$$

and finger positions $(.020833, -.020833)$ m. T4 additionally stores a temporary preparation-target pose with $x, y \sim U[-.03, .03]$ and yaw $U[0, 2\pi]$, distinct from the release state.

The evaluator source stores recipes in `configs/scenarios/` and task bindings in `configs/tasks/`; the generation entry point is `se3_wam_benchmark/cli/generate_scenarios.py`. The evaluation payload contains the concrete `evaluation_reference/scenes/validation.jsonl` and fourteen `evaluation_reference/scenes/suites/sr_<task>.yaml` files. Each suite uses one repeat and filters the shared manifest by task. These files, with the stored scenario IDs and source seeds, define the replay set used for the paired comparisons.

The accompanying local material package at `results/replay_materials/` contains the archived evaluator, task and sampling configurations, all accepted scenarios, and both ordered selections. Portable copies change only execution paths; originals are retained. The archived manifest parser reads all 1,400 scenarios without changing their scientific fields, and the task/configuration loaders read all fourteen definitions. The supplied checker also verifies disjoint partitions, identities, seeds, and asset references. These checks parse the materials; physics and policy replay require the additional runtime and model assets listed in the package instructions.

Table S2. Historical validation proposal ranges. Length, linear velocity, angle, and angular velocity use m, m/s, rad, and rad/s unless marked otherwise. Positions are in world coordinates except the stated ramp-local offsets. Continuous intervals are uniform proposals; discrete choices are equiprobable before feasibility checks. Evaluation replays the stored accepted scenarios, whose distribution need not remain uniform. Harmonic centers are not initial positions.

| Task | Initial-state proposal | Motion or event parameters |
|---|---|---|
| grasp | Harmonic center $c_x \in [-.08, .08]$, $c_y \in [-.26, -.19]$, $z = .822$; yaw $[-1.1, 1.1]$. | $v_y \in [.025, .125]$, $A_x \in [.015, .035]$, $f_x \in [.3, .7]$ Hz; $\omega_z \in [-1.2, 1.2]$. |
| t1_belt | Target $x \in [-.08, .08]$, $y \in [-.26, -.18]$, $z = .822$; identity orientation, zero initial twist. | Belt $v_y \in [.04, .10]$. |
| t1_occ | Same target proposal as Belt; canopy at $(.322, -.29, 1.20)$. | Same belt range; clearance parameter .20; physical occlusion. |
| t1_xyz | Harmonic center $c_x \in [-.025, .025]$, $c_y \in [-.24, -.20]$, $z = .818$; yaw $[-.35, .35]$. | $v_y \in [.04, .09]$, $A_x \in [0, .03]$, $f_x \in [.25, .65]$ Hz; $\omega_z = 0$. |
| t1_so3 | Same as XYZ. | Same translation; $\omega_z \in [-1.2, -.4] \cup [.4, 1.2]$, equal sign probabilities. |
| t2_trans | Goal $x \in [-.13, -.08]$, $y \in [-.18, -.12]$, $z = .825$; fixed orientation. | Goal $v_y \in [.03, .06]$; no rotation. |
| t2_se3 | Same goal $x, y$ range, $z = .804$; yaw $[-.3, .3]$. | Goal $v_y \in [.03, .06]$; $\omega_z \in [-.25, -.08] \cup [.08, .25]$, equal sign probabilities. |
| t3_phase | Wand $x, y \in [-.03, .03]$, $z = .870$, yaw $[0, 2\pi]$; wire phase $[-.25, .35]$. | Wire center $(.10, -.02, .94)$; world-X rotation $\omega_x = -\{2.5, 3, \ldots, 7.5\}^\circ$/s (11 equal-probability values). |
| t3_full | Wand at $(-.05, -.02, 1.04)$ with identity orientation; same wire phase range. | Same wire rotation; cradle and release requirement enabled. |
| t4_sphere | Ramp-local offsets: lateral $[-.015, .015]$, longitudinal/normal $(-.145, .030)$; lip $(.06, -.02, 1.10)$. | Gravity-driven from rest; gate release at tick 0; slope $[5, 15]^\circ$; target/ramp sliding friction $[.7, .9]$; unsupported capture. |
| t4_sphere_tabletop | Lateral $[-.015, .015]$, longitudinal/normal $(-.124, .034)$; lip $(0, -.06, .808)$. | Same slope and release; target/ramp friction $[.7, .9]$, table friction .35. |
| t4_slider | Lateral $[-.015, .015]$, longitudinal/normal $(-.129, .024)$; supported ramp. | Same slope and release; target/ramp friction $[.025, .045]$, table friction $[.012, .018]$. |
| t4_can | Lateral $[-.010, .010]$, ramp-axis yaw $[-4, 4]^\circ$; longitudinal position follows gate geometry and oriented can extent. | Same slope and release; target/ramp friction $[.25, .45]$, table friction $[.09, .15]$. |
| t5_replan | $x \in [-.005, .005]$, $y = -.29$, $z = .812$; yaw $[-.08, .08]$; zero initial twist. | Belt $v_y \in [.12, .14]$; equal-probability left/right striker triggered at $y = -.13$; stroke .098, pulse .35 s, actuator scale $[1, 1.1]$, force cap 4 N; command delay 50 ms. |

## A.3. Partitions and Statistical Interpretation

The 140-scene development set has ten scenes per task. Its disjoint complement has ninety per task, totaling 1,260. The future screen reports the two partitions separately. The system-reference study also keeps them separate. The combination screen explicitly selects on all 1,400 scenes while retaining both component counts. Thus its selection score includes development data, and the 1,260 complement is now part of the project's development evidence. Neither membership disjointness nor a different rollout seed makes it a final blind test.

The future study uses each condition's own fourth-epoch endpoint and one training seed. Across arms, the scene, source seed, and inference seed are paired. There are 18,900 accepted outcomes, 1,260 per arm; missing scenes and infrastructure-error exclusions are zero. Inference seeds span 20261058–20262317 for the designated selection indices. Matching an inference seed specifies rollout randomness but does not imply that unlike training graphs consumed identical random-number sequences.

For two policies on the same scenes, let $b$ count candidate successes/reference failures and $c$ candidate failures/reference successes. Then

$$\Delta\mathrm{SR} = 100(b - c)/1260 \tag{S1}$$

in percentage points. The report uses the exact two-sided McNemar test on discordant pairs. Fourteen comparisons to P00 and thirteen to P03 form the twenty-seven-comparison family; adjusted values use $\min(1, 27p)$. The 378 task-level comparisons remain exploratory.

We additionally compute 95% pointwise percentile intervals by resampling paired scenes within each task. For every replicate, each task retains ninety pairs (one hundred in the combination study), and the fourteen task differences are averaged with equal weights. Sampling a multinomial from each task's four paired outcome counts is equivalent to resampling its binary scene pairs. We use 100,000 replicates and the 2.5th/97.5th percentiles with linear interpolation. The generator is NumPy 2.3.5 PCG64; seeds are 20261017 plus the zero-based lexicographic comparison index for the twenty-seven future contrasts, and 20261017/20261018 for I versus F/E. The post-hoc I–E analysis excluding phase passage retains thirteen tasks with seed 20261019. Tasks are processed in lexicographic order. These intervals condition on the trained checkpoints and fixed task mixture; they do not include training-seed variation or candidate-selection uncertainty. They are not simultaneous intervals; the corrected McNemar family above remains the basis for the overall future-screen comparisons.

### A.4. Inputs and Time

The completed future screen uses one current fixed-camera RGB view, exact depth/camera geometry, robot-state history, and privileged object information. A simulator field stored beside an image is not automatically a deployable sensor observation. Future target states and label-validity masks supervise their respective branches and are excluded from the action condition. A current-object-pose oracle and a future oracle are distinct information conditions.

Physical high-level steps are separated by 50 ms. All one hundred T5 scenarios use a one-step command delay (50 ms); the other thirteen tasks use zero. The native action queue returns a zero command until the delay elapses, then applies the oldest queued command. Forecast labels cover 50–400 ms relative to the source observation. Synchronous evaluation pauses the environment for policy computation, so it cannot measure the consequence of wall-clock observation age. Later fixed-delay and real-delay evaluations must retain their own time traces and report whether physics continues while the policy computes. Future labels follow the demonstration's recorded geometry under its executed actions. The current validity masks check geometric support, projection, and visibility. They do not establish that the same post-contact future remains valid for an alternative candidate action. Action-consistent label support is **[pending for online generation]**.

## B. Reference Implementation and Budgets

### B.1. Actual Representation-Screen Configuration

The reference D model uses `se3wam_v2/split_ffn`, four transformer layers, width 768, eight attention heads, and factorized attention. DINOv3 ViT-L/16 is frozen [8]. A $224 \times 224$ agent-view image produces 196 current visual tokens. H0V has zero visual-history query tokens. The information track also supplies exact depth and camera geometry. Projected image patches receive a view embedding, a linear embedding of mean valid metric depth, and a linear embedding of the patch-center robot-base XYZ before the view interface. All reported internal conditions use one agent view, so this interface returns that view directly and does not perform cross-view attention. XYZ is backprojected from fixed source pixels (indices 7, 23, …, 215) using camera intrinsics and extrinsics. These coordinates also enter 3-D RoPE for visual Q/K and the corresponding flow queries; object, robot, meta, and action tokens receive no XYZ. Robot input retains eight EEF7 states at 20 Hz. Eight object tokens encode current GT pose and seven motion intervals using 25 numeric statistics and pose/interval/acceleration validity flags.

The 25 object dimensions are $(p, \log R, \Delta p, \log \Delta R, \Delta t, v, \omega, a, \alpha)$, with dimensions $(3, 3, 3, 3, 1, 3, 3, 3, 3)$. The eight frames span $-350$ to 0 ms. Slots 0–6 encode adjacent intervals at their centers ($-325$ to $-25$ ms); only slot 7 retains the current absolute pose. Velocity divides interval displacement by elapsed time, and acceleration differences adjacent velocities at their interval centers. No state preceding the window is used. Each dimension is standardized by training statistics and invalid dimensions are zeroed. Three validity flags are appended before a $28 \to 768 \to 768$ MLP with GELU and a time embedding. Eight object, eight EEF, and 196 visual tokens form the scene bank. Sixteen learned meta tokens are separate inputs. The object-motion values come from simulator truth in this track. The current-pose history track removes historical object poses and all derived interval kinematics while preserving the current pose and robot history. This is a different information condition, not a different name for the same input.

D shares attention QKV/output projections and splits feed-forward networks by world/meta, flow, and action stream. Scene tokens access scene tokens; meta tokens access scene and meta; flow accesses scene and flow; action accesses meta and action. Consequently, flow supervision can change shared weights without an online flow-to-action path. The deterministic pose head pools sixteen final meta tokens.

### B.2. Coordinates, Losses, and Generation

Target future translation is cumulative displacement in the fixed metric frame. Rotation is the left-relative vector $\phi_h = \mathrm{Log}(R_{t+\delta_h} R_t^\top)^\vee$. For valid targets $\mathcal{V}$, the pose loss is

$$\mathcal{L}_{\mathrm{pose}} = \frac{1}{|\mathcal{V}|} \sum_{h \in \mathcal{V}} \big[ \|\hat{d}_h - d_h\|_2^2 + 0.1\, d_{\mathrm{SO}(3)}(\hat{R}_{t+\delta_h}, R_{t+\delta_h})^2 \big]. \tag{S2}$$

The external weight multiplying this loss is separate from the internal rotation coefficient. A full twist exponential would couple translation to rotation through its Jacobian and is not the implemented parameterization.

The action has eight E7 steps: base-frame translation increment, left-relative rotation vector, and gripper command. Translation and rotation training bounds are 0.05 m and 0.5 rad per axis. The inherited evaluation translation limit is 0.10 m per axis. Training normalization uses the mean and population standard deviation of valid, non-repeated issued E7 commands in the training split, accumulated in FP64 and stored in FP32. Action flow matching uses a unit Gaussian in these coordinates. Each query samples $u \sim U[0, 1)$, shared across its full $8 \times 7$ action chunk. Deployment reverses normalization, decodes rotation vectors, and clips physical translation/rotation at 0.10 m/0.5 rad per axis; the gripper command uses its sign. Scene flow uses a metric Gaussian source with standard deviation 0.02 m. Deterministic auxiliary readouts, including pose, do not consume flow-matching source noise and time.

Only the action generator runs during Q2 deployment, using five Euler steps. Conditioning caching is on; temporal/velocity caching is off. Each replan executes one command. Trainable initialization is fresh for seed 0 rather than inherited from a trained common-reference checkpoint. Global batch is 16. Four epochs produce 40,104 updates and 641,664 chunks. AdamW uses learning rate $2 \times 10^{-4}$, betas $(0.9, 0.95)$, $\epsilon = 10^{-8}$, weight decay 0.05, and gradient-norm clipping at 1. After 1,000 warm-up updates, cosine decay spans the four-epoch schedule with a minimum learning-rate ratio of 0.1. Training uses FP32 with TF32 enabled, zero dropout, and an FFN expansion ratio of four. P03 weights action, scene flow, and pose losses by one; Alpha is disabled. For cumulative scene flow $F_h$, the dynamic weight is $w_h = 1 + 128\,\mathrm{clip}(\|F_h - F_{h-1}\|_2/c, 0, 1)$, where $F_0 = 0$ and $c = 0.014620669186115265$ m. The weighted FM loss divides by three times the valid correspondence count, rather than by the sum of weights. P00, P02, and P05 finish with an eight-device topology; other arms finish with sixteen. Some retained runs have recorded topology transitions. All endpoints retain the declared exposure; topology transitions can change the subsequent numerical training trajectory.

## B.3. Alternative Target Definitions

Table S3 specifies the target arrays after omitting batch and query axes. Unless stated otherwise, the eight horizons are 50, 100, …, 400 ms. Surface correspondence follows fixed source geometry; a future occlusion does not by itself invalidate that identity. Track visibility is a separate label, using depth consistency within $\max(0.001\text{ m}, 0.001z)$ in the camera frame.

Wan receives nine RGB frames at $192 \times 160$ normalized to $[-1, 1]$. Its first temporal latent is dropped; the remaining two encode frames 1–4 and 5–8. Codec labels are extracted from actual decoder motion vectors. Target-track sampling starts nearest the mask centroid, then selects the farthest point; row-major order resolves ties. Fewer than 32 points are zero-padded and masked. P08/P09 adapt mechanisms rather than reproduce an entire published model.

Regression objectives average squared error over channels and valid positions. Tracks add mean two-coordinate MSE and visibility binary cross-entropy with logits on valid correspondences. Valid positions are pooled across the batch, rather than averaging episode means. Native pose uses the geometric loss above. Replacement losses P04–P09, native flow, native pose, and action each have outer weight one.

Aalpha adds motion-magnitude MSE with weight 0.5 while preserving P03's Alpha-head initialization. P12, P13, and P14 add future depth, spatial DINO, and dynamic-region RGB supervision, each with weight 0.1. P15 combines the three losses at these same weights. A non-existent head has an inapplicable readout, not a zero error. The combination cannot identify every pairwise interaction. Raw total losses across these objectives are not commensurate.

## B.4. System-Reference Training

ACT and Diffusion Policy train from scratch for 50,000 updates at global batch 32. The $\pi_{0.5}$ adaptation starts from `pi05_base` and trains 30,000 updates at global batch 256 [7]. DynamicVLA uses 50,000 updates at global batch 44, initialized from `hzxie/dynamic-vla-DOM` with vision, text, and connector frozen. It uses observations at indices $[-2, 0]$, a 20-step horizon, and normalization fitted on 800 demonstrations. OpenWAM uses a fresh fine-tuning run from the official Alpha foundation, for five epochs, 34,490 updates at global batch 192, with its declared foundation components. DynamicWAM uses its official full initialization and 40,000 updates at global batch 128 [6, 9]. DreamVLA's 20-epoch and 40-epoch schedules use independent training runs, 48,220 and 96,440 updates at global batch 512; the shorter endpoint is not the midpoint checkpoint of the longer schedule [10]. DreamVLA receives seven causal RGB frames, EEF position/Euler orientation, two finger positions, and fixed task text. Its missing wrist view is black; inference uses DDIM10 and classifier-free guidance 1.5. OpenWAM receives current RGB, seven joint values, measured gripper opening, and task text. One agent view and two black wrist views form a $384 \times 320$ composite; state is mapped to its native 80-dimensional slots. Ten synchronous joint inference steps produce nine video frames and 32 actions, with video decoding disabled.

Executors also differ: ACT uses temporal aggregation, Diffusion Policy executes eight actions, $\pi_{0.5}$ ten, DynamicWAM sixteen, and OpenWAM thirty-two before replanning. DreamVLA predicts three steps but replans every tick, aggregates the latest three forecasts with exponential

Table S3. Future targets in the representation screen. Shape excludes batch/query axes. Displacements are cumulative from the current state except codec vectors, which retain the decoder's native correspondence. All targets come from the recorded demonstration.

| Target | Shape | Support, units, and validity |
|---|---|---|
| Scene flow | $196 \times 8 \times 3$ | Fixed $14 \times 14$ patch centers; base-frame meters; valid source depth/geometry and finite future displacement. |
| Target pose | $8 \times 6$ | Cumulative translation (m) and left-relative rotvec (rad); valid object pose and query. |
| P04: Wan | $48 \times 2 \times 12 \times 10$ | Frozen Wan2.2 VAE posterior mean and native normalization; future groups end at 200/400 ms; valid query. |
| P05: dense 2-D | $8 \times 2 \times 224 \times 224$ | Fixed source pixels; displacement divided by 224; valid source and finite future projection. |
| P06: codec | $8 \times 2 \times 64 \times 64$ | Each future/current pair is encoded as P/I frames; native vectors divided by 64; actual valid vector regions. |
| P07: scene tracks | $196 \times 8 \times 3$ | Same patch source points; two displacement channels divided by 224 and one visibility logit; valid correspondences. |
| P08: object DINO | $8 \times 1024$ | Object-mask-area-weighted patch pooling; channel layer normalization; positive mask area. |
| P09: object tracks | $32 \times 8 \times 3$ | Deterministic farthest-point sampling in the current target mask; displacement and visibility as P07; padded points invalid. |
| $A_\alpha$: magnitude | $196 \times 8$ | Cumulative flow norm (m); same correspondence mask as flow. |
| P12: depth | $8 \times 1 \times 224 \times 224$ | Metric future depth; finite positive pixels. |
| P13: DINO | $8 \times 196 \times 1024$ | Frozen DINOv3-L/16 patches, channel layer normalization; valid query/horizon/patch. |
| P14: dynamic RGB | $8 \times 3 \times 224 \times 224$ | RGB divided by 255; visible surfaces moving more than 1 mm in world coordinates, including robot and other geometry. |

Table S4. Training exposure and development counts for all system-reference endpoints. Batch is global; all endpoints use training seed 0. Initialization and execution are part of each adapted system, not matched factors. The two DreamVLA endpoints are independent training schedules. The 1,260-scene counts appear in the main system table.

| System | Updates | Batch | Success /140 | Execution / initialization note |
|---|---|---|---|---|
| ACT | 50,000 | 32 | 50 | Scratch; temporal aggregation |
| Diffusion Policy | 50,000 | 32 | 54 | Scratch; execute 8 actions |
| $\pi_{0.5}$ | 30,000 | 256 | 89 | `pi05_base`; execute 10 |
| DynamicVLA | 50,000 | 44 | 24 | Official DOM checkpoint; native continuous pipeline |
| OpenWAM-Alpha | 34,490 | 192 | 76 | Official Alpha foundation; execute 32 |
| DynamicWAM | 40,000 | 128 | 72 | Official full initialization; execute 16 |
| DreamVLA: 20 epochs | 48,220 | 512 | 112 | Independent schedule; aggregate forecasts, execute 1 |
| 40 epochs | 96,440 | 512 | 114 | Independent longer schedule |
| Study P00 | 40,104 | 16 | 88 | Fresh task network; frozen DINO-L; execute 1 |
| Study P03 | 40,104 | 16 | 93 | Same reference recipe as P00 |
| Study I | 40,104 | 16 | 96 | Fresh CNN condition; full visual history; execute 1 |

weights $\exp(-0.01k)$ over their age order, and executes one command. The internal representation screen executes one. System-level interpretation retains these differences. Matched-budget comparisons must also report each system's inputs, initialization, and execution configuration.

# C. History Encoding and Conditional Information

## C.1. Completed Conditions

The current-pose study compares eight history encodings plus a fresh GT-history reference. Every visual-history condition preserves current GT target pose and eight robot-state samples, while historical object pose and derived kinematics are masked. Its reference future supervision is held fixed. All eight current-pose conditions are complete: current image, Window, Codec-MV, Flow, Flow12D, PACT-Age, Wan, and PACT-Uniform. The fresh GT-history reference also has complete training, 1,400-scene evaluation, and probe reports. It retains current RGB and adds privileged object-motion history. PACT-Age distributes fifty-six history queries from older to newer frames as $[2, 4, 6, 8, 10, 12, 14]$; PACT-Uniform uses seven groups of eight. These adapt PACT-WAM's temporal allocation principle [5] through per-

frame query compression, rather than reproducing its complete model.

All windows contain eight samples at 20 Hz, spanning $-350$ to 0 ms. Current/Window/PACT add fixed 768-dimensional sinusoidal frame-time embeddings in seconds; time subtraction is performed in FP64 before conversion to FP32. Window retains every past patch. PACT uses one shared eight-query bank; age allocation linearly interpolates that bank with aligned endpoints, rather than adding frame-specific query parameters. Layer-normalized multi-head attention produces query-residual outputs.

Flow is bidirectional Farnebäck flow [2] on area-resized $64 \times 64$ RGB (pyramid scale 0.5, three levels, window 15, three iterations, polynomial neighborhood 5, sigma 1.2, flags 0). The four intervals last 100, 100, 100, and 50 ms. Validity requires finite vectors, in-frame endpoints, valid bilinear reverse sampling, and

$$\|f + b(x+f)\|^2 \le 0.01(\|f\|^2 + \|b(x+f)\|^2) + 0.5. \quad \text{(S3)}$$

Flow color uses direction as hue, unit saturation, and magnitude divided by the interval's 99th percentile, clipped to $[0, 1]$, as value. Invalid pixels are black with validity zero. The four-channel field enters a $4 \to 768$ convolution with kernel/stride 8, followed by layer normalization, 64 learned spatial embeddings, and fourteen shared queries with interval-time embeddings. Attention, residual addition, and layer normalization produce fourteen tokens per interval.

For valid flow $f = (d_x, d_y)$, let $r = \|f\|_2$, bars denote pixel means, and $q$ be the 99th percentile of $r$. Flow12D supplies

$$(\bar{d}_x, \bar{d}_y, \bar{r}, q, \Delta t, \bar{d}_x/\Delta t, \bar{d}_y/\Delta t, \bar{r}/\Delta t, q/\Delta t, a_x, a_y, \|a\|_2).$$

Acceleration differences adjacent mean velocities over interval-center time gaps of 100, 100, and 75 ms; the first interval has no acceleration. All dimensions except $\Delta t$ are divided by 64, with no fitted mean/standard deviation. Twelve feature-valid flags form a 24-dimensional input to a $24 \to 768 \to 768$ GELU MLP. A separate $2 \to 768 \to 768$ MLP embeds the left/right times relative to the current frame. Pure Flow zeros the numeric values and flags but retains these four slots and their time embeddings. Thus the comparison adds magnitude, velocity, and acceleration statistics to a color encoding that normalizes magnitude.

Codec labels come from independent MPEG-4 Part 2 I/P pairs with no B frames, GOP 2, quantizer 2, YUV420p, one thread, and bitrate $10^6$. Negated decoder backward vectors remain attached to destination blocks, unlike source-grid optical flow. Intra, uncovered, and out-of-frame-reference regions are invalid; no forward/backward confidence filter is applied. Wan uses a frozen FP32 Wan2.2 VAE posterior mean after right-padding T8 with the current frame. Its temporal groups end at $-350, -150, 0$ ms, with real-frame fractions $1, 1, 0.75$. A $48 \to 768$ kernel/stride-2 convolution creates 147 source tokens; learned spatial embeddings and an MLP for group time/fraction precede 56-query compression. VAE caches are cleared per window.

The original implementation allocates 1,584 scene slots for Current/Window, 272 for Flow/Flow12D, and 268 for the other compressed conditions. These counts include masked past-object slots; sixteen meta tokens are additional. Allocated slots and valid information are reported separately.

Each of the nine conditions has probe extraction over 350 episodes and 1,050 windows with 306 valid readouts. A completed CPU audit checks 9,450 saved NPZ files through 385,350 array comparisons: shared recorded inputs, labels, initial flow-matching noise, and the frozen current-image, geometry, robot-state, and clock baseline match across conditions. Oracle inputs match across the eight current-pose conditions; the intended GT-motion difference is excluded from equality checks. The completed point estimates give no visual-history gain over the current-image control. The nine full result files match on task, scene, source seed, and inference seed. Against Current, gain/loss counts are 137/223 for Window, 133/148 for Codec-MV, 176/204 for Flow, 183/208 for Flow12D, 132/246 for PACT-Age, 194/232 for Wan, 122/200 for PACT-Uniform, and 238/162 for GT history. The main-paper intervals use Appendix A's task-stratified bootstrap with 100 paired scenes per task and 100,000 replicates. Seeds are 20261017–20261024 in this order. They describe scene uncertainty for the E4 seed-0 endpoints on the development pool, without training-seed, selection, or multiplicity adjustment. Outcome pairing and the separate probe-array audit cover distinct experimental records.

GT history has 586 joint successes and 414 joint failures with Current; its net gain is 76 scenes, or 5.43 points (95% interval 3.07–7.79). Table S7 shows a pronounced task tradeoff: XYZ grasping gains forty successes, while phase passage falls from 84/100 to 1/100. The aggregate score therefore does not imply a uniform gain from motion input.

## C.2. Recipe Differences and Required Contrasts

This study uses frozen DINO-L, retains the original Alpha objective, and has a cosine schedule over the first 10,026 updates followed by three epochs at $2 \times 10^{-5}$. The combination study uses DINO-S or CNN, disables Alpha, and applies cosine scheduling across all 40,104 updates. Its stronger score therefore cannot be subtracted from the history-study score to estimate a visual-history effect. Flow12D and Wan train fresh on sixteen PRO6000 devices; PACT-Uniform trains on eight PRO6000 devices, with a same-topology restart from update 781. Window, Codec-MV, and Flow resume sixteen-device A100 prefixes on PRO6000; PACT-Age additionally changes from sixteen to eight devices. All preserve global batch 16 and the four-epoch exposure. The device histories preclude a matched throughput comparison. GT history

Table S5. History encoders in the current-pose study. All conditions preserve the current 196 DINO-L visual tokens, T8 EEF states, and current object pose. History tokens are width 768 and carry no XYZ. Counts below concern added visual/numeric history tokens, not allocated scene slots.

| Condition | Sampled times (ms) | History encoding | Added tokens |
|---|---|---|---|
| Current | 0 | Past image slots are masked. | 0 |
| Window | $-350{:}50{:}0$ | Seven past frames, 196 DINO patches each; no query compression. | 1,372 |
| Flow | $-350, -250, -150, -50, 0$ | Four $2 \times 64 \times 64$ flows; each RGB+validity field yields 64 patches and 14 learned-query outputs. | $56 + 4$ |
| Flow12D | Same as Flow | Same visual encoder; four numeric slots receive twelve motion statistics and their flags. | $56 + 4$ |
| Codec-MV | Same as Flow | Four actual codec MV fields, encoded by the same RGB+validity backend; no numeric slots. | 56 |
| Wan | $-350{:}50{:}0$ | Repeat current frame once; $48 \times 3 \times 14 \times 14$ VAE latent $\rightarrow$ 147 patches $\rightarrow$ 56 queries. | 56 |
| PACT-Uniform | $-350{:}50{:} - 50$ | Shared eight-query bank reads each of seven past frames. | 56 |
| PACT-Age | Same as Uniform | Shared query bank interpolated to $[2, 4, 6, 8, 10, 12, 14]$ per frame. | 56 |

Table S6. Complete history results. The total is a development summary, not a separate held-out policy test.

| Condition | /140 | /1,260 | /1,400 | SR (%) |
|---|---|---|---|---|
| Current image | 80 | 668 | 748 | 53.43 |
| Window | 69 | 593 | 662 | 47.29 |
| Codec-MV | 77 | 656 | 733 | 52.36 |
| Flow | 76 | 644 | 720 | 51.43 |
| Flow12D | 75 | 648 | 723 | 51.64 |
| PACT-Age | 65 | 569 | 634 | 45.29 |
| Wan | 70 | 640 | 710 | 50.71 |
| PACT-Uniform | 68 | 602 | 670 | 47.86 |
| *Reference with privileged target history* | | | | |
| Fresh GT history | 84 | 740 | 824 | 58.86 |

runs on eight PRO6000 devices until update 18,060, then resumes on sixteen with a native RNG reset. Its completed endpoint has 40,104 updates and 641,664 sampled chunks. This topology change is included in the interpretation of the single-training-seed comparison.

The planned core comparison crosses current-pose versus GT-motion input with current-image versus visual-history encoding under a common recipe. Multiple training seeds and paired scenes are needed to estimate the conditional history benefit. A motion probe and an action intervention answer different questions: recoverability of velocity does not demonstrate that the policy used velocity to improve interaction.

### C.3. Offline History Perturbations

For each of fourteen tasks, the probe protocol assigns 15/5/5 distinct episodes to training/validation/test, with three windows per episode. Table S8 uses the resulting 70 test episodes and 210 windows, with initial-noise seed 17. This is a probe split within the 350 policy-validation episodes, not a new policy test set. The intervention repeats the current image in all seven past slots or reverses the seven past images. Depth and calibration follow the same image permutation; the current frame, object information, robot history, timestamps, and masks remain fixed. These perturbed histories need not be physically realizable.

Each intervention re-encodes the window without a previous perception cache, rebuilds the condition, and reuses the actual Action/Flow initial-noise arrays. The native reports accept 840 same-input/noise controls per condition. Repeated-baseline and identity controls have zero differences; both current-image references have zero action and readout changes under every history intervention. Separate three-step checks in the eight current-pose conditions confirm exact condition-cache reconstruction and zero influence of masked old GT states. The GT-history reference intentionally consumes those states and is excluded from that masking check. Future labels are read only after prediction.

All seven history encodings change the issued first action under these perturbations, despite their lack of a success-rate gain in the main screen. For example, Window translation changes by 9.83 mm under repetition and 1.92 mm under reversal; Wan changes by 20.05 and 24.25 mm. The comparison distinguishes sensitivity to history from task benefit. It does not measure a closed-loop intervention or assign a mechanism to the success-rate differences.

This analysis aggregates the original per-task, episode-averaged reports. Their sample identities and source reports match the native records; the separate array audit above verifies the shared inputs and initial noise.

Table S7. History-study successes, one hundred scenes per task and condition. The first eight conditions retain current GT target pose; GT motion retains current RGB and adds privileged target-motion history. All use one training seed and the fourth-epoch endpoint; the 1,400 scenes are the development pool.

| Task | Current | Window | Codec | Flow | Flow12D | PACT-Age | Wan | Uniform | GT motion |
|---|---|---|---|---|---|---|---|---|---|
| Grasp | 66 | 71 | 62 | 65 | 66 | 62 | 62 | 52 | 88 |
| Belt | 87 | 48 | 89 | 99 | 94 | 65 | 88 | 61 | 95 |
| Occlusion | 72 | 67 | 62 | 74 | 70 | 59 | 87 | 70 | 94 |
| SO(3) | 42 | 36 | 34 | 38 | 44 | 29 | 37 | 17 | 56 |
| XYZ | 41 | 48 | 54 | 53 | 47 | 55 | 62 | 44 | 81 |
| SE(3) placement | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Trans. placement | 13 | 6 | 15 | 13 | 14 | 12 | 40 | 16 | 10 |
| Full passage | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| Phase passage | 84 | 6 | 99 | 0 | 0 | 28 | 3 | 67 | 1 |
| Can | 70 | 96 | 82 | 79 | 92 | 79 | 53 | 68 | 100 |
| Slider | 84 | 80 | 70 | 80 | 85 | 59 | 73 | 70 | 78 |
| Sphere | 41 | 55 | 46 | 65 | 76 | 38 | 63 | 45 | 66 |
| Tabletop sphere | 96 | 97 | 98 | 100 | 100 | 77 | 100 | 96 | 99 |
| Replan | 52 | 52 | 22 | 54 | 35 | 71 | 42 | 63 | 56 |
| Total / 1,400 | 748 | 662 | 733 | 720 | 723 | 634 | 710 | 670 | 824 |

Table S8. Offline sensitivity in the eight current-pose conditions. Mean change in the issued first action, relative to the unchanged input, over 70 probe-test episodes (210 windows, fourteen tasks). Values are equal-task means of episode means. Translation is L2 change in mm; rotation is geodesic change in degrees. These are action changes, not action errors or success gains.

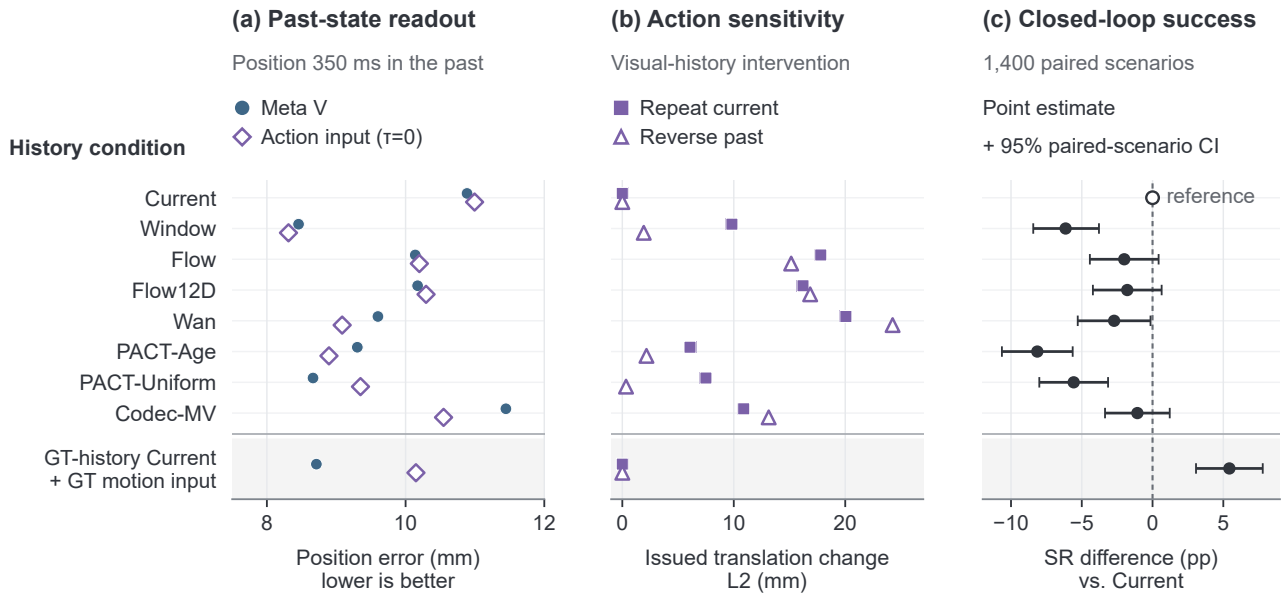
| | Repeat current | | Reverse past | |
|---|---|---|---|---|
| Condition | mm | deg | mm | deg |
| Current image | 0.00 | 0.00 | 0.00 | 0.00 |
| Window | 9.83 | 1.28 | 1.92 | 0.34 |
| Codec-MV | 10.89 | 1.59 | 13.13 | 2.00 |
| Flow | 17.80 | 2.20 | 15.15 | 2.51 |
| Flow12D | 16.21 | 2.43 | 16.85 | 2.95 |
| PACT-Age | 6.07 | 0.48 | 2.17 | 0.25 |
| Wan | 20.05 | 2.70 | 24.25 | 3.98 |
| PACT-Uniform | 7.50 | 0.67 | 0.33 | 0.04 |

## C.4. Frozen Readouts and Conditional Residuals

The readouts test which recorded motion variables can be decoded from frozen features. They use the same 15/5/5 episode split above, selected by sorted native episode identity. Ridge regression is fitted independently for each task, feature site, and label. The penalty is chosen from $\{0.1, 1, 10, 100\}$ by validation coordinate MSE, without refitting on validation or test data. Features use training-only means and population standard deviations, with values at most $10^{-8}$ replaced by one; targets are centered but not scaled, and the intercept is unpenalized. A shuffled control permutes the forty-five training-window labels with seed 1729, leaving validation and test labels unchanged.

**Features and targets.** The history site captures the visual-history encoder output entering the first Transformer layer: Window pools spatially within each past frame, while compressed encoders retain ordered tokens. The other sites are final Meta V and the input immediately before the action head at the first Action-branch function evaluation, $\tau = 0$, each mean/max pooled to 1,536 dimensions. The 18,048-dimensional conditioned baseline contains frozen current DINO-L features pooled to $4 \times 4$, current depth and validity, robot-base patch coordinates and validity, eight EEF states, relative times and masks, and the condition's actual oracle values and validity. Thus GT history supplies additional motion information despite equal baseline width. Past-position labels refer to the target body origin in the robot-base frame at lags of 50, 200, and 350 ms. Velocity uses the last 50 ms backward difference; the future label is the demonstrated 400 ms continuation displacement.

**Readout comparisons.** A retained readout uses site features alone; a joint readout concatenates them with the conditioned baseline. Joint gain is baseline physical error minus joint physical error. For conditional residuals, the baseline penalty is selected on validation data, then five episode-disjoint folds produce training residuals: twelve episodes fit each baseline and three provide its held-out targets. Site features predict these residuals. Validation and test residuals instead use the baseline fitted on all fifteen training episodes. Reconstruction adds this fixed baseline prediction to the predicted residual. This tests a restricted decoder, not all




Seed 0. Offline: 14-task means, no CIs. Closed loop: scenario uncertainty only.

Figure S2. History decodability, action sensitivity, and closed-loop success. (a) Frozen ridge error for position 350 ms in the past, from final Meta V or the first action-head input ($\tau = 0$). (b) L2 change of the first issued translation after action limits, under repeat-current or reverse-past intervention with identical initial noise. Panels (a,b) show descriptive means across fourteen fixed tasks, without confidence intervals. (c) Success-rate difference from Current over 1,400 paired scenarios, with pointwise 95% intervals from 100,000 task-stratified resamples. These intervals condition on single-seed checkpoints and the development pool. The separated GT-history reference receives additional GT motion. Both current-image references have no visual history to perturb; GT motion remains intact. Lower readout error and nonzero action sensitivity do not establish a task benefit.

information available to the policy.

**Aggregation and uncertainty.** Position error is the per-window Euclidean distance, averaged over three windows per episode, five test episodes per task, and fourteen equally weighted tasks. It is not RMSE or the model's pose-head error. Translation and velocity metrics cover all fourteen tasks; rotation and angular velocity cover only the three orientation-observable tasks. Per-task intervals resample the five episode means 2,000 times with seed 1729; they condition on the fitted probes and policy endpoints. Macro results average task point estimates and have no confidence intervals.

**Results.** Window has lower 350 ms position-error point estimates than Current at both Meta V (8.46 versus 10.89 mm) and the action input (8.31 versus 10.99 mm), while its success rate is 6.14 points lower (Fig. S2). Readable past state and sensitivity to image history therefore coexist with worse task performance in this comparison. The conditioned residuals give a similarly limited result: across three sites, nine conditions, and three labels—350 ms past position, last-50 ms velocity, and 400 ms future displacement—74 of 75 defined task-macro $R^2$ values are negative; the exception is Flow12D Meta V velocity at 0.034. Residual $R^2$ uses the test-residual mean as its reference, and its macro averages defined task scores rather than pooling tasks. A negative value does not establish absence of encoded information. Current pose is oracle-determined in every condition; GT history also determines past pose and velocity. Their unique-information residual scores are undefined, not zero, and baseline approximation errors are reported separately. The six GT-history past-position/velocity cells are therefore excluded from the count above. These readouts do not establish that the conditioned input is an information-theoretic sufficient statistic.

## C.5. Matched-Current and Occlusion Diagnostics

History substitutions require matching the current target pose, robot state, view, and task condition while changing valid preceding motion. The matching tolerance, group counts, and history-necessary subset acceptance remain **[pending]**. A substitute that changes the current state tests an additional intervention. Occlusion tests must report visibility and duration in physical time, while sampling-gap tests

preserve the meaning of displacement by retaining elapsed time. The planned outputs link motion decoding, action change, contact outcome, and unconditional task success. No-contact failures remain in the success denominator rather than receiving zero contact error.

# D. Future Supervision and Combination Selection

## D.1. All Fifteen Future Conditions

All conditions in Table S9 have action supervision. Scene-flow coverage includes target regions, so P01 is not a pure context-only condition. P03 versus P01/P02 compares target-pose and scene-flow objectives under the common action path; it does not isolate an exclusive target/context spatial partition. Main-paper Fig. 3 visualizes all fifteen point estimates with task-stratified paired bootstrap intervals, as specified in Appendix A.

P00/P01/P02/P03 test no future objective, scene flow, pose, and their combination. P04–P07 replace the scene target while retaining pose. P08/P09 replace the object target while retaining scene flow. Aalpha and P12–P15 add supervision to P03. Full target definitions and source units are given in Appendix B. None of these conditions changes deployment into online forecast consumption.

## D.2. Paired Outcomes and Task Heterogeneity

Table S10 gives gains, losses, difference intervals, and adjusted exact McNemar probabilities against both reference conditions. P05's adjusted probabilities are 0.00844 versus P00 and 0.01877 versus P03. The remaining positive overall differences are not established advantages. The absence of a detected difference is not an equivalence test.

Table S11 gives every task count with equal denominators. The strongest task-specific changes are exploratory. For P05 versus P03, rolling-sphere success falls from 50 to 23 while `tray_contact` failures rise from 40 to 67. Conversely, post-impact replanning rises from 54 to 60. P09's gains in translational placement and replanning coexist with fewer orientation-sensitive grasps. These observations narrow the conditions to investigate, without treating a native failure code as a learned-mechanism explanation.

The lowest count of native safety failures is P06's 250, but its 96 timeouts are the highest. P15 has only 37 timeouts and 343 safety failures. A favorable change in one category must be interpreted alongside all outcomes. Every policy failure remains in the 1,260-scene denominator.

## D.3. Combination Selection

The next screen crosses three future choices, two visual encoders, and current-target RGB reconstruction on/off. It preserves a full visual window and privileged target-motion input, disables Alpha and late injection, and trains each condition fresh for four epochs with the full-run cosine schedule. Its twelve candidates and the selected configuration's matched P00 and P03 controls are complete. All fourteen conditions start with sixteen devices and global batch 16. L and N resume on eight devices with batch 2 per device, preserving the optimizer, schedule, normalization, sample cursor, and 40,104-update budget, but resetting destination-rank random states. K resumes on a different GPU type with sixteen devices. M completes training on sixteen devices and moves only its remaining 1,260-scene evaluation to eight. I, E, and F retain sixteen-device training. The full grid thus matches nominal exposure, but its P09 candidates do not share one hardware and random-state history.

**CNN and temporal conditioning.** The trainable CNN has four convolution–GroupNorm–GELU blocks with channels $3 \to 32 \to 64 \to 128 \to 192$. Each convolution has kernel 4, stride 2, padding 1, and no bias; GroupNorm uses eight groups. The $224 \times 224$ image becomes $14 \times 14$ patches, followed by token LayerNorm ($\epsilon = 10^{-6}$) and a $192 \to 768$ projection. RGB is divided by 255, bilinearly resized with antialiasing, and normalized by means $(0.485, 0.456, 0.406)$ and standard deviations $(0.229, 0.224, 0.225)$. Geometry embeddings follow Appendix B. The full T8 window retains $8 \times 196 = 1,568$ visual tokens, eight EEF tokens, and eight object-kinematic tokens: 1,584 scene tokens when the full window is available, plus sixteen learned meta tokens. Relative-time embeddings and trunk attention fuse the window; there is no history-query compression or temporal mean pooling.

**Current-target reconstruction.** The mean of the final sixteen meta tokens enters a $768 \to 256 \to 3,072$ MLP with GELU and final sigmoid, producing $3 \times 32 \times 32$ RGB. Simulator target geometry IDs select the currently visible target and its tight bounding box. The crop is resized to $32 \times 32$ (bilinear RGB, nearest-neighbor mask); values outside the mask are zero. Loss is masked L1 with outer weight 0.1, divided by the global count of valid RGB elements. An empty mask contributes zero without dropping the training sample. The reconstruction is a training readout, not an action input. I, E, and F share this path, the same D backbone, and the same training normalization from 1,324,374 valid issued actions in 11,200 training episodes. The 350 validation episodes do not contribute normalization statistics.

Candidate I obtains 935/1,400 by the predefined selection score. Against J, the same P07/CNN condition without reconstruction, the paired counts are 817 joint successes, 118 I-only successes, 79 J-only successes, and 386 joint failures. Its net 39-scene gain is distributed unevenly over tasks, as Table S13 shows. The selected candidate is carried forward for testing across training seeds.

Table S9. All fifteen future-supervision conditions, grouped by the objective changed. Each uses one training seed, four epochs, common privileged object-motion input, and action-only deployment. The 140 development scenes and 1,260-scene complement remain separate. Scene flow includes target regions.

| | | Development | Disjoint complement | |
|---|---|---|---|---|
| ID | Future supervision | Success / 140 | Success / 1,260 | SR (%) |
| *Reference and component objectives* | | | | |
| P00 | No future supervision | 88 | 791 | 62.78 |
| P01 | 3-D scene flow | 95 | 782 | 62.06 |
| P02 | Target pose | 92 | 789 | 62.62 |
| P03 | Scene flow + target pose | 93 | 786 | 62.38 |
| *Scene replacements: target-pose objective retained* | | | | |
| P04 | Wan latent + pose | 92 | 783 | 62.14 |
| P05 | Dense 2-D displacement + pose | 87 | 746 | 59.21 |
| P06 | Codec motion + pose | 86 | 795 | 63.10 |
| P07 | Sparse scene tracks + pose | 90 | 807 | 64.05 |
| *Target replacements: scene-flow objective retained* | | | | |
| P08 | Scene flow + object DINO | 89 | 794 | 63.02 |
| P09 | Scene flow + target tracks | 88 | 808 | 64.13 |
| *Additional objectives: scene flow + target pose retained* | | | | |
| $A_\alpha$ | P03 + motion magnitude | 89 | 778 | 61.75 |
| P12 | P03 + depth | 90 | 791 | 62.78 |
| P13 | P03 + DINO | 92 | 788 | 62.54 |
| P14 | P03 + dynamic RGB | 89 | 790 | 62.70 |
| P15 | P03 + depth / DINO / dynamic RGB | 94 | 802 | 63.65 |

Table S10. Paired outcomes over the same 1,260 scenes. Gain means candidate success/reference failure; loss means candidate failure/reference success. Intervals are pointwise task-stratified paired bootstrap intervals for the SR difference, conditional on the trained checkpoints (Appendix A). Exact two-sided McNemar tests use the full family of fourteen P00 and thirteen P03 contrasts: $p_{\mathrm{adj}} = \min(1, 27p)$. P03 against itself is inapplicable.

| | P00: no future supervision | | | | P03: scene flow + target pose | | | |
|---|---|---|---|---|---|---|---|---|
| ID | Gain | Loss | 95% CI (pp) | Adjusted $p$ | Gain | Loss | 95% CI (pp) | Adjusted $p$ |
| *Reference and component objectives* | | | | | | | | |
| P01 | 80 | 89 | $[-2.70, 1.27]$ | 1 | 72 | 76 | $[-2.14, 1.51]$ | 1 |
| P02 | 61 | 63 | $[-1.83, 1.51]$ | 1 | 61 | 58 | $[-1.43, 1.90]$ | 1 |
| P03 | 64 | 69 | $[-2.14, 1.35]$ | 1 | — | — | — | — |
| *Scene replacements: target-pose objective retained* | | | | | | | | |
| P04 | 74 | 82 | $[-2.54, 1.27]$ | 1 | 71 | 74 | $[-2.06, 1.59]$ | 1 |
| P05 | 53 | 98 | $[-5.40, -1.75]$ | 0.00844 | 47 | 87 | $[-4.92, -1.43]$ | 0.0188 |
| P06 | 81 | 77 | $[-1.59, 2.22]$ | 1 | 68 | 59 | $[-1.03, 2.46]$ | 1 |
| P07 | 85 | 69 | $[-0.63, 3.17]$ | 1 | 81 | 60 | $[-0.16, 3.49]$ | 1 |
| *Target replacements: scene-flow objective retained* | | | | | | | | |
| P08 | 75 | 72 | $[-1.59, 2.06]$ | 1 | 59 | 51 | $[-0.95, 2.22]$ | 1 |
| P09 | 91 | 74 | $[-0.56, 3.25]$ | 1 | 79 | 57 | $[0.00, 3.49]$ | 1 |
| *Additional objectives: scene flow + target pose retained* | | | | | | | | |
| $A_\alpha$ | 60 | 73 | $[-2.78, 0.71]$ | 1 | 53 | 61 | $[-2.30, 1.03]$ | 1 |
| P12 | 73 | 73 | $[-1.83, 1.83]$ | 1 | 63 | 58 | $[-1.27, 2.06]$ | 1 |
| P13 | 70 | 73 | $[-2.06, 1.59]$ | 1 | 60 | 58 | $[-1.51, 1.83]$ | 1 |
| P14 | 71 | 72 | $[-1.90, 1.75]$ | 1 | 62 | 58 | $[-1.35, 2.06]$ | 1 |
| P15 | 74 | 63 | $[-0.95, 2.70]$ | 1 | 63 | 47 | $[-0.32, 2.94]$ | 1 |

Table S11. Future-study successes: ninety scenes per task and 1,260 per condition. Columns follow the task families in Table S1; all tasks are retained, including zero-success tasks. Occl. denotes physical occlusion; Trans. denotes translating-receptacle placement; Tabletop denotes tabletop sphere capture.

| | Reference | T1: target motion | | | | T2: placement | | T3: passage | | T4: capture | | | | T5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ID | Grasp | Belt | Occl. | SO(3) | XYZ | SE(3) | Trans. | Full | Phase | Can | Slider | Sphere | Tabletop | Replan | Total |
| *Reference and component objectives* | | | | | | | | | | | | | | | |
| P00 | 85 | 90 | 90 | 75 | 76 | 0 | 8 | 3 | 15 | 90 | 69 | 45 | 86 | 59 | 791 |
| P01 | 84 | 90 | 90 | 64 | 69 | 0 | 18 | 0 | 11 | 90 | 74 | 46 | 90 | 56 | 782 |
| P02 | 87 | 90 | 90 | 78 | 73 | 0 | 7 | 0 | 0 | 90 | 71 | 59 | 89 | 55 | 789 |
| P03 | 86 | 90 | 88 | 75 | 77 | 0 | 7 | 0 | 6 | 86 | 80 | 50 | 87 | 54 | 786 |
| *Scene replacements: target-pose objective retained* | | | | | | | | | | | | | | | |
| P04 | 83 | 85 | 84 | 73 | 76 | 0 | 20 | 0 | 0 | 90 | 72 | 54 | 89 | 57 | 783 |
| P05 | 84 | 89 | 90 | 66 | 75 | 0 | 10 | 0 | 0 | 90 | 71 | 23 | 88 | 60 | 746 |
| P06 | 86 | 88 | 89 | 79 | 70 | 0 | 6 | 4 | 2 | 89 | 72 | 60 | 85 | 65 | 795 |
| P07 | 82 | 90 | 90 | 78 | 71 | 0 | 17 | 0 | 0 | 89 | 79 | 55 | 90 | 66 | 807 |
| *Target replacements: scene-flow objective retained* | | | | | | | | | | | | | | | |
| P08 | 84 | 90 | 90 | 71 | 81 | 0 | 13 | 0 | 0 | 90 | 77 | 60 | 90 | 48 | 794 |
| P09 | 84 | 90 | 89 | 65 | 78 | 0 | 21 | 0 | 0 | 90 | 77 | 59 | 90 | 65 | 808 |
| *Additional objectives: scene flow + target pose retained* | | | | | | | | | | | | | | | |
| $A_\alpha$ | 86 | 90 | 90 | 72 | 75 | 0 | 3 | 0 | 0 | 85 | 73 | 55 | 90 | 59 | 778 |
| P12 | 83 | 90 | 90 | 76 | 76 | 0 | 5 | 0 | 0 | 89 | 77 | 57 | 90 | 58 | 791 |
| P13 | 86 | 90 | 90 | 80 | 73 | 0 | 6 | 0 | 0 | 88 | 78 | 54 | 90 | 53 | 788 |
| P14 | 85 | 89 | 90 | 72 | 79 | 0 | 5 | 0 | 0 | 90 | 80 | 53 | 88 | 59 | 790 |
| P15 | 87 | 90 | 90 | 79 | 81 | 0 | 11 | 0 | 0 | 90 | 75 | 54 | 90 | 55 | 802 |

Table S12. Native evaluator failure counts out of 1,260 scenes per condition, in the same design groups as the success tables. Safety is the evaluator category, not a certified safety measure. Every failure remains in the success-rate denominator.

| ID | Future supervision | Boundary | Safety | Task viol. | Timeout | Total |
|---|---|---|---|---|---|---|
| *Reference and component objectives* | | | | | | |
| P00 | No future supervision | 84 | 297 | 17 | 71 | 469 |
| P01 | 3-D scene flow | 62 | 324 | 23 | 69 | 478 |
| P02 | Target pose | 77 | 331 | 17 | 46 | 471 |
| P03 | Scene flow + target pose | 76 | 295 | 51 | 52 | 474 |
| *Scene replacements: target-pose objective retained* | | | | | | |
| P04 | Wan latent + pose | 79 | 310 | 41 | 47 | 477 |
| P05 | Dense 2-D displacement + pose | 87 | 356 | 14 | 57 | 514 |
| P06 | Codec motion + pose | 96 | 250 | 23 | 96 | 465 |
| P07 | Sparse scene tracks + pose | 47 | 329 | 20 | 57 | 453 |
| *Target replacements: scene-flow objective retained* | | | | | | |
| P08 | Scene flow + object DINO | 61 | 322 | 42 | 41 | 466 |
| P09 | Scene flow + target tracks | 67 | 299 | 35 | 51 | 452 |
| *Additional objectives: scene flow + target pose retained* | | | | | | |
| $A_\alpha$ | P03 + motion magnitude | 57 | 315 | 65 | 45 | 482 |
| P12 | P03 + depth | 62 | 327 | 40 | 40 | 469 |
| P13 | P03 + DINO | 71 | 326 | 27 | 48 | 472 |
| P14 | P03 + dynamic RGB | 70 | 322 | 18 | 60 | 470 |
| P15 | P03 + depth / DINO / dynamic RGB | 59 | 343 | 19 | 37 | 458 |

Table S13. Combination-study successes, grouped by future target and visual encoder. DINO is frozen DINO-S; CNN is trained from scratch. Minus/plus denotes current-target RGB reconstruction off/on; parentheses retain source condition IDs. E and F are the selected configuration's matched scene-flow/pose and no-future controls. Each task has one hundred scenes, including development scenes, and each condition has one training seed. Totals use the predefined 1,400-scene selection pool.

| | P15: auxiliary | | | | P07: scene tracks | | | | P09: target tracks | | | | P03 | P00 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Task | DINO − (B) | DINO + (A) | CNN − (D) | CNN + (C) | DINO − (H) | DINO + (G) | CNN − (J) | CNN + (I) | DINO − (L) | DINO + (K) | CNN − (N) | CNN + (M) | CNN + (E) | CNN + (F) |
| *Reference grasp* | | | | | | | | | | | | | | |
| Grasp | 90 | 89 | 91 | 92 | 90 | 92 | 87 | 90 | 79 | 85 | 77 | 87 | 85 | 78 |
| *T1: target motion* | | | | | | | | | | | | | | |
| Belt | 100 | 100 | 99 | 100 | 100 | 100 | 100 | 97 | 99 | 100 | 90 | 99 | 100 | 100 |
| Occlusion | 100 | 100 | 99 | 99 | 99 | 100 | 100 | 97 | 96 | 99 | 82 | 98 | 99 | 98 |
| SO(3) | 86 | 71 | 80 | 88 | 75 | 87 | 69 | 86 | 69 | 62 | 64 | 78 | 78 | 71 |
| XYZ | 58 | 58 | 82 | 75 | 73 | 75 | 89 | 82 | 44 | 56 | 82 | 84 | 78 | 82 |
| *T2: receptacle motion* | | | | | | | | | | | | | | |
| SE(3) | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Translation | 8 | 4 | 6 | 5 | 9 | 1 | 14 | 4 | 4 | 5 | 9 | 14 | 9 | 11 |
| *T3: constrained passage* | | | | | | | | | | | | | | |
| Full | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Phase | 0 | 0 | 0 | 0 | 0 | 0 | 28 | 72 | 0 | 0 | 0 | 2 | 0 | 0 |
| *T4: object capture* | | | | | | | | | | | | | | |
| Can | 99 | 99 | 99 | 98 | 100 | 99 | 99 | 100 | 97 | 99 | 96 | 99 | 99 | 99 |
| Slider | 88 | 85 | 88 | 91 | 74 | 83 | 86 | 81 | 84 | 84 | 88 | 88 | 84 | 77 |
| Sphere | 71 | 67 | 67 | 63 | 47 | 40 | 53 | 56 | 68 | 68 | 62 | 64 | 68 | 45 |
| Tabletop sphere | 100 | 100 | 100 | 99 | 99 | 100 | 95 | 98 | 100 | 100 | 100 | 100 | 99 | 99 |
| *T5: post-impact replanning* | | | | | | | | | | | | | | |
| Replan | 73 | 63 | 64 | 57 | 74 | 72 | 76 | 72 | 80 | 72 | 79 | 57 | 62 | 50 |
| Total / 1,400 | 873 | 836 | 875 | 867 | 840 | 849 | 896 | 935 | 820 | 830 | 829 | 870 | 861 | 810 |

The matched P00 control F retains I's trainable CNN, full visual window, privileged motion input, and current-target masked RGB reconstruction (weight 0.1), with future objectives disabled. Both use fresh seed 0, global batch 16, four epochs, and the full-run cosine schedule. F obtains 79/140 and 731/1,260 successes, totaling 810/1,400. For I versus F, the paired counts are 746 joint successes, 189 I-only successes, 64 F-only successes, and 401 joint failures. The 125-scene net gain is 8.93 percentage points (95% pointwise interval 7.00–10.86). It includes 72 on phase passage and 22 on replanning; translational placement loses seven. These are results on the pool used to select I; the interval does not correct for candidate selection.

The matched P03 control E changes the future objectives to scene flow and target pose. It obtains 87/140 and 774/1,260 successes, totaling 861/1,400. I versus E has 786 joint successes, 149 I-only successes, 75 E-only successes, and 390 joint failures. The net 74-scene gain is 5.29 percentage points (95% pointwise interval 3.50–7.07). Phase passage contributes 72 of these net successes; the remaining thirteen tasks contribute two in total. The aggregate difference is therefore highly task-concentrated within this development pool. A post-hoc calculation excluding phase passage gives 863/1,300 for I and 861/1,300 for E: 0.15 points (interval $[-1.69, 2.00]$). This descriptive sensitivity analysis does not replace the predefined fourteen-task score. Confirmation requires unused scenes and additional training seeds.

### D.4. Granularity and Prediction-to-Action Evidence

A future comparison must separate role coverage, variable choice, spatial allocation, and compute. The fifteen-target screen changes target packages, not only allocation at equal budget. A direct fine-target/coarse-context test still requires a compatible uniform reference with the same identity information, horizon support, and declared token/compute budget. Its result is **[pending controlled granularity comparison]**.

Physical readouts should compare target translation and rotation at each horizon; flow and tracks require valid correspondences and visibility. Their association with contact error is a separate analysis. The accepted source contains

readout arrays, but this manuscript does not substitute flow-matching validation loss for freely generated physical error. The aligned prediction/contact figure requires **[generated predictions, actual states, and contact events]**. If a future oracle is introduced, it must use the same action interface and target times. Any oracle-gap ratio is reported only with a supported, non-negligible denominator.

# E. Backbone, Information Flow, and Cross-Checks

## E.1. D/E Backbone Comparison

The planned backbone comparison uses the confirmed complete history/future configuration, retaining the reference $G_0$ program where action does not read predicted future. D shares QKV and output projections across WM, Flow, and Action, with a separate feed-forward network for each. Adapted E assigns QKV/output projections, adaptive layer normalization (AdaLN), and feed-forward networks to two parameter groups: World and Action. Scene, meta, and any active future-Flow tokens share World parameters; Action has its own group. World observation/meta tokens use diffusion time $\tau = 0$, while Flow keeps its per-token time. For a future target decoded without scene-Flow flow matching, both candidates omit the active Flow branch.

The comparison also retains each candidate's internal normalization: E's World AdaLN at $\tau = 0$ has randomly initialized scale/shift biases, whereas D's WM LayerNorm starts with unit affine scale and zero shift. Thus, the shared input normalization accompanies a comparison of parameter organization and internal normalization, rather than isolating feed-forward sharing.

The planned two-arm comparison trains fresh seeds 0, 1, and 2, with the same initialization sources, data exposure, optimizer schedule, and endpoint. For each seed, common input projections, history/time parameters, and readouts start identically; the backbone parameter groups are initialized separately. Both arms require fresh training under this paired initialization. If the current four-epoch recipe is retained, there are 240,624 updates across six runs and 8,400 evaluation episodes at 1,400 per endpoint. These are planned counts. The active-backbone parameter difference is targeted within 5%; frozen frontend parameters and their runtime costs are recorded separately. Matching exposure or parameter count does not match training FLOPs.

The predeclared practical SR and non-inferiority margins are two percentage points, with a 10% p95 latency benefit considered for cost selection. Intervals must support the corresponding superiority, equivalence, or non-inferiority criterion before that label is used. If three seeds are inconclusive, both arms may be expanded to five under the existing plan. Same-device public-policy timing includes internal perception/action computation and excludes external rendering and oracle access. Selection and uncertainty are **[pending]**.

## E.2. Seven Information-Flow Conditions

R0 versus R1 tests auxiliary supervision under action-only deployment. R1's two deployment modes test the cost and effect of generating an unused future from the same checkpoint. R1–R4 examine information access. R2 versus R5 and R3 versus R6 compare synchronous and sequential programs. Sequential second stages train on clean first-stage labels and deploy on first-stage predictions. The corresponding train/test mismatch belongs in the interpretation.

The current generative state is scene flow with shape $[B, S, 196, 8, 3]$; pose, tracks, depth, DINO, and RGB can be deterministic auxiliary readouts. A selected sparse-track or Wan representation requires its own generated-state, normalization, noise, and loss definition before this experiment can run. Replacing it with scene flow would change the question. This interface is **[pending final representation binding]**.

The planned three seeds yield twenty-one trained configurations and twenty-four deployment configurations, including R1's second mode; at 1,400 scenes each, that is 33,600 rollout episodes. Alpha is fixed to zero. Per-modality Euler5 is common, but joint inference uses five combined calls and sequential inference uses five plus five. Actual calls, latency, parameters, training exposure, and valid readouts are reported alongside SR.

## E.3. Mechanisms and Final Design

An independently replaceable future interface permits oracle, valid matched-current swap, time-shift, or calibrated-noise diagnostics. Each intervention holds perception, robot state, and action-source noise fixed. Changes in action alone show sensitivity; evidence for useful forecast use also requires task and safety outcomes. Removing a branch at inference and retraining without its loss answer different questions.

After selecting information flow, the study revisits retained future representations and applies the selected flow to the unselected backbone. History choices are revisited when the future target changes. Only the necessary comparisons are repeated; previously valid matched endpoints can be reused. Ranking reversals change the conditional design choice. Multi-seed and scenes not used for selection are required before passing a complete $(H, Z, B, G)$ design to pretraining. The selected configuration, cross-check estimates, and mechanism readouts remain **[pending]**.

Table S14. Planned Q3 conditions; no measured result is assigned to these rows. F denotes a generated future state and A an action state. Arrows specify information access, independently of whether the future loss is enabled. R1 has two deployments of the same trained checkpoint. Sequential second stages use clean ground truth in training and generated first-stage values at test time.

| ID | Future loss | $F \to A$ | $A \to F$ | Cross-stream training input | Inference order |
|---|---|---|---|---|---|
| R0 | No | — | — | None | A only |
| R1 | Yes | — | — | None | A only |
| R1 | Yes | — | — | None | Joint F and A |
| R2 | Yes | Yes | — | Synchronous noisy states | Joint F and A |
| R3 | Yes | — | Yes | Synchronous noisy states | Joint F and A |
| R4 | Yes | Yes | Yes | Synchronous noisy states | Joint F and A |
| R5 | Yes | Yes | — | Clean F for second-stage A | F, then A |
| R6 | Yes | — | Yes | Clean A for second-stage F | A, then F |

# F. Pretraining, Cost, and External Evaluation

## F.1. Measured Policy Cost and Physical Delay

The future-screen policy call means span 177.071–184.212 ms; p95 spans 178.726–185.133 ms; peak allocated memory spans 1.571–1.617 GiB. These are same-device, fixed-order measurements including internal perception and action computation. Loading, external oracle reads, I/O, rendering, and priming are excluded. Each arm uses three recorded eight-frame sequences, six warm-up calls, and thirty measured calls (ten per sequence). After each reset, seven real calls prime the observation history; the eighth is timed with CUDA synchronization before and after the call. The sequences cover translation, occlusion, and post-impact replanning. Inference is FP32 with TF32 enabled, batch/sequence dimensions $B = S = 1$, and five Euler action steps. NVML identifies the shared device as `NVIDIA RPBZZZ6`, with 102,641,958,912 bytes of memory; CPU affinity is 0–2. Memory is the post-warm-up PyTorch allocation, including resident weights. One measurement round supports this range; differences of a few milliseconds do not establish a robust speed ranking.

For deployed commands, action age is the difference between sensor capture and command application. The planned deployment profile separates capture, availability, preprocessing, inference, queuing, issue, and application. It records hardware, precision, batch size, warm-up, synchronization, and timed sample counts. Fixed-delay simulation tests a controlled lag; wall-clock execution tests the actual combined system. The final same-device profiles, deadline misses, and action-age distributions are **[pending]**.

## F.2. Initialization Controls

Three predefined controls separate initialization from downstream design. For DreamVLA, the control removes the LIBERO policy checkpoint while retaining the MAE/CLIP initialization, teacher targets, losses, and the independent twenty-epoch recipe (48,220 updates, global batch 512). For $\pi_{0.5}$, the control replaces the base initialization with the LIBERO expert while retaining the 30,000-update, global-batch-256 downstream recipe. These comparisons use the original fourteen-task evaluation partitions.

A separate transfer study uses an action-only version of the four-layer, width-768 D trunk with a trainable CNN, T8 RGB and robot-state history. It removes target-pose input, depth/camera geometry, the flow stream, and all future and target-region reconstruction objectives. The two downstream conditions share this network and normalization fitted on the same 11,200 demonstrations; one starts fresh and the other loads LIBERO-90 visual-motor pretraining [4]. The pretraining protocol uses 4,050 demonstrations, 450 held-out demonstrations, thirty epochs, and 8,520 updates at global batch 256. Both downstream runs reset optimizer and scheduler state and use four epochs, 40,104 updates, and global batch 16. This contrast tests visual-motor transfer within the simplified policy; it is neither a P07 pretraining ablation nor a language-conditioned LIBERO evaluation. Endpoint success rates for these three controls are **[pending]**. They do not supply the formal robot-data base described below.

## F.3. Recipe and Robot-Data Base

After the design cross-checks, the study fixes input modalities, history support, future targets/horizons, information flow, and action semantics. The next stage optimizes the existing loss weights and training recipe, and changes depth/width within the chosen architecture for scale comparison. Changing the representation or adding an information path belongs to a new design comparison rather than an optimizer ablation. Appendix H reports a completed recipe comparison on development combinations G and I; it does not replace the planned study of the final selected design.

DROID supplies real robot demonstrations [3]. Preparation checks sensor timing, calibration, depth, command versus measured motion, gripper semantics, and training-only normalization. It does not provide object pose GT; the target-pose pipeline must be separately validated for any selected

pose-based objective. Only targets used by the selected design and necessary ablations need to be produced. The data report includes valid-label masks and rejected-action rates. This stage is **[planned]**.

The formal base is trained on real robot data. Two independent post-training branches start from the same base: one for simulation and one for real-robot tasks. Simulation-from-scratch experiments supply recipe experience and a strong reference, while their weights do not initialize the formal robot-data base or the real-robot branch. Comparisons with and without pretraining share downstream data and exposure, and additionally allow equally budgeted recipe searches. They report the pretraining cost explicitly. Data, model, and compute scale have separate axes.

### F.4. Planned External and Real-Robot Evidence

Single-arm DOMINO [1] tests dynamics under a different task/data protocol. Static compatibility uses an explicitly selected native suite such as LIBERO [4]. Real conveyor manipulation evaluates perception, prediction, action, and timing together. The final phase reuses completed pretraining, scaling, and independent adaptation results, then fills uncovered object, scene, motion, and delay conditions. It does not repeat an already completed scaling matrix.

The real-robot protocol must specify objects, speed/delay strata, trial allocation, ordering, calibration, safety termination, and human intervention. External native scores are reported separately from the fourteen-task simulation SR. Final training scale, robot trials, and real-world performance are **[pending]**.

## G. Diagnostics and Qualitative Evidence

### G.1. Supervision at the Executed Action Position

All 800 historical full-passage demonstrations release only at the final action: chunk position $h_7$ has release supervision, while the executed position $h_0$ has none. This identifies an action-position coverage gap.

Tail-window training produces 2/16 successes in diagnostic validation and 1/6 in a selected-scene follow-up. For phase passage, three seeds after 256 updates give 0/96, 12/96, and 26/96; at 1,024 updates all three give 0/96. Six first-action-weighting endpoints total 576 failures in 576 trials, and six GT-role diagnostic endpoints also total zero successes in 576. These local budgets are separate from the fourteen-task study.

A zero action-source deployment intervention changes the three 256-update scores from 0/12/26 to 0/1/46 successes out of 96 each. This post-hoc intervention is exploratory. A reproducibility investigation links some same-seed trajectory branching to one-gray-level RGB changes; fixed full-input/RNG inference and fixed-action replay remain consistent.

### G.2. Gripper Execution Diagnostics

The gripper investigation compares sign decoding, a fixed 0.1 dead zone, and nearest-label ternary projection and history-hold mappings. The ternary projection maps to $\{-1, 0, +1\}$ with thresholds at $\pm 0.5$. For the reported fifth-epoch endpoint, the 1,260-scene counts are 698 for sign, 742 for the fixed dead zone, 749 for nearest-label ternary projection, and 741 for history hold. The ternary projection does not show a clear overall improvement over the fixed dead zone; history hold adds failures outside the capture subset. Neither is adopted as the production default in the evidence cited here.

The broader fourth-to-eighth-epoch dead-zone report has not passed strict historical trajectory-consistency checks and is excluded from main-table claims. Decoder effects and representation effects require separate comparisons. Appendix H reports the subsequent frozen-head study and its failed regression gate.

### G.3. Offline Target-Pose Predictions

Figure S3 compares deterministic fourth-epoch P15 pose readouts with two kinematic baselines on 84 states (six per task). All eight horizons are valid, yielding 672 state–horizon pairs. Quantiles use linear interpolation with equal state weights; states from one episode may be dependent. Median translation error rises from 1.45 mm at 50 ms to 4.04 mm at 400 ms; the corresponding rotation medians are $0.47°$ and $1.07°$. The cases are selected by mean translation error, independently of task success.

**Kinematic baselines.** CV and CA use the final two and three observed GT poses, respectively, within the same history available to P15. They do not read future labels or instantaneous simulator velocities. The saved raw positions, xyzw quaternions, and FP64 timestamps verify a 50 ms interval $d$ for all states. The resulting last-interval derivatives agree with the model's unstandardized object-input values. For positions $p_{-2}, p_{-1}, p_0$, CV predicts $h(p_0 - p_{-1})/d$. CA predicts $hv_0 + h^2 a/2$, where

$$v_0 = \frac{3p_0 - 4p_{-1} + p_{-2}}{2d}, \qquad a = \frac{p_0 - 2p_{-1} + p_{-2}}{d^2}.$$

CV retains the final interval velocity; CA extrapolates it from the $-25$ ms interval center to the current time.

In the fixed robot-base frame, define $\omega_+ = \mathrm{Log}(R_0 R_{-1}^\top)^\vee/d$ and $\omega_- = \mathrm{Log}(R_{-1} R_{-2}^\top)^\vee/d$. Rotational CV uses $\Delta R(h) = \mathrm{Exp}(h\omega_+^\wedge)$. CA sets $\omega_0 = 1.5\omega_+ - 0.5\omega_-$ and $\alpha = (\omega_+ - \omega_-)/d$, and integrates

$$\dot{\Delta R}(t) = [\omega_0 + t\alpha]_\times \Delta R(t), \qquad \Delta R(0) = I.$$

Exponential midpoint steps are at most 1 ms; halving the step changes any predicted rotation by at most $0.000548°$. There is no smoothing, clipping, or fitted baseline parameter.

Table S15. DEV84 offline prediction errors: 84 states and eight horizons, yielding 672 dependent observations. CV/CA use the last two/three GT poses; P15 also uses the full history and its other observation inputs.

| | Translation (mm) | | Rotation (°) | |
|---|---|---|---|---|
| Method | Mean | Median | Mean | Median |
| P15 | 3.762 | 2.771 | 1.772 | 0.857 |
| CV | 12.943 | 1.444 | 9.675 | 0.175 |
| CA | 22.413 | 0.634 | 10.114 | 0.170 |

Table S15 reports both mean and median errors. P15 has lower means but higher pooled medians than either extrapolator: the baselines combine many small errors with a large-error tail. Using each state's mean over eight horizons, P15 has lower translation error than CV and CA on 42/84 and 37/84 states, and lower rotation error on 27/84 and 34/84. Thus the aggregate means do not establish uniform prediction superiority. The comparison characterizes offline error distributions under GT motion inputs.

### G.4. Closed-Loop Alignment

The planned comparison will align observations, predictions, actual futures, action application, and contact in physical time. It will include P05's adverse sphere condition, a sparse-trajectory condition, and a matched future-to-action intervention, recording scene/camera identity, units, visibility, outcomes, and selection rules. Synchronized rollout exports, real-robot and forecast-intervention videos are **[pending]**.

## H. Additional Training and Execution Studies

### H.1. Paired Training-Recipe Study

The recipe study tests two development combinations, G (frozen DINOv3-S) and I (trainable CNN), both with P07 supervision, the full visual window, and current-target RGB reconstruction. For each profile, training seeds 0 and 1 compare global batch 16 on sixteen GPUs against batch 64 on thirty-two GPUs. The larger-batch recipe doubles the learning rate from $2\times10^{-4}$ to $4\times10^{-4}$ and reduces warm-up from 1,000 to 250 updates, preserving 16,000 warm-up samples. This is a joint recipe and resource comparison, not an isolated batch-size intervention.

All eight runs complete four epochs and 641,664 samples. B16 uses 40,104 updates; B64 uses 10,028, retaining each epoch's 32-sample tail without padding, repetition, or dropping. AdamW, clipping, FP32/TF32, losses, normalization, and the four-epoch cosine exposure are held fixed. All final checkpoints undergo the same sixteen-device full350 validation and the original 1,400-scene evaluation with paired scene and inference seeds. The seed-0 B16 runs are fresh reruns: their scene-level success outcomes exactly reproduce the corresponding BASE2 endpoints. They do not add a new independent training seed. Seed 1 supplies the additional realization.

**Performance and variability.** For G, B64 minus B16 gives $+0.43$ and $-0.86$ percentage points at seeds 0 and 1, with paired 95% intervals $[-1.21, 2.07]$ and $[-2.50, 0.79]$. For I, the changes reverse direction: $-3.64$ $[-5.64, -1.64]$ and $+5.57$ $[3.57, 7.57]$ points. At B16, I falls from 935 to 857 successes between seeds; phase passage falls from 72 to 14. Neither the original seed-0 score nor the mean response to a recipe change establishes stability. The matched E/F controls have not been repeated at seed 1.

The two-seed mean recipe differences are $-0.21$ $[-1.36, 0.96]$ points for G and $+0.96$ $[-0.50, 2.39]$ for I. Intervals use 20,000 percentile bootstrap samples, RNG seed 20261017, resampling 100 scene identities within each task and retaining their batch and training-seed pairing. They condition on these two trained seeds and fourteen tasks; they do not estimate variation over new training seeds. Table S17 also shows task losses hidden by aggregate scores: G loses 17 XYZ successes at seed 0 and 15 SO(3) successes at seed 1. Auxiliary metrics are not uniformly preserved; for example, I's pose rotation error increases by 6.17% and 14.35%.

**Time and adoption.** Mean job-creation-to-result time decreases from 302.43 to 120.77 minutes for G and from 257.81 to 101.38 minutes for I. These measured times include queueing, setup, training, saving, validation, export, and evaluation. They do not identify a hardware-only speedup. Training-compute GPU-hours decrease from 70.22 to 52.47 for G and from 59.42 to 42.31 for I; this measure excludes saving and validation and is not a billing record. The predeclared working tolerance was a three-point aggregate SR loss. G satisfies it for both observed seeds, whereas I does not. The project therefore retains B16 as the general recipe; B64 is a conditional option for this G configuration when aggregate SR is the criterion. It is not adopted for I, task-sensitive comparisons, or an untested final design.

### H.2. Frozen Three-State Gripper Readout

The execution study freezes the fifth-epoch P03 policy and learns a three-state readout. Its 889-dimensional input concatenates the current action hidden state at the final Euler evaluation ($\tau = 0.8$), eight frames of available end-effector pose, gripper opening and validity, the sampled six-dimensional arm-action horizon, and the current raw gripper value. A 128-hidden-unit GELU MLP predicts open/hold/close logits; argmax replaces only the executed slot-0 gripper command. A second head with the same hidden width reads only the raw gripper value. No contact truth or future label is added to the observation. The frozen policy,

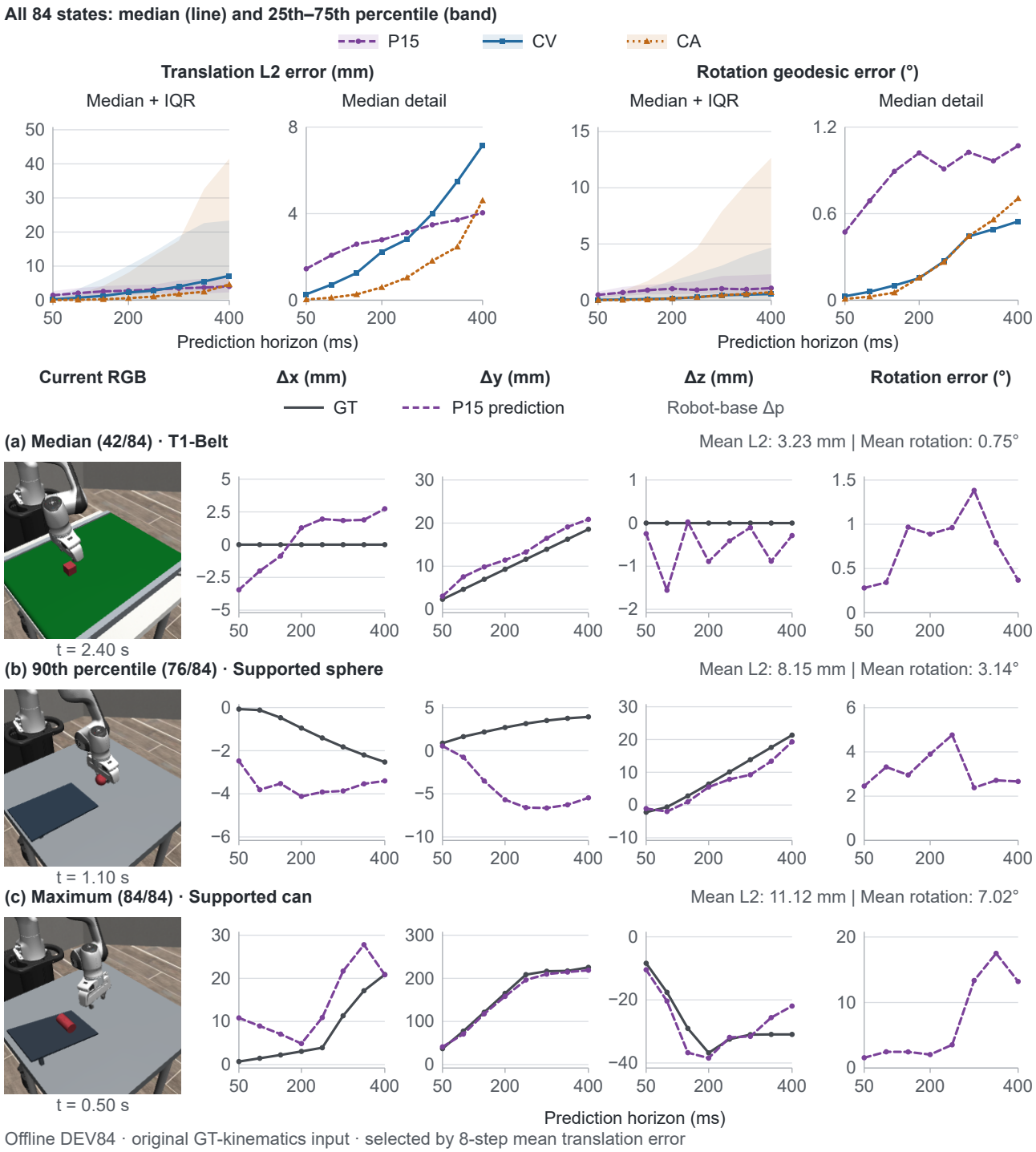



Figure S3. Offline P15 predictions and kinematic extrapolation. The DEV84 probe set contains 84 query states from 28 episodes across fourteen tasks. Top: P15, constant-velocity (CV), and constant-acceleration (CA) median translation L2 and rotation geodesic errors at each horizon; bands show the 25th–75th percentiles across states, not confidence intervals. The adjacent Median detail panels show the same median curves on expanded linear scales, without IQR bands. Bottom: we sort states by their mean translation L2 error over eight horizons and select nearest ranks 42, 76, and 84 (50th, 90th, and 100th percentiles), breaking ties by state index. Each row shows the original current image, predicted and true cumulative translation in robot-base coordinates, and rotation geodesic error at 50–400 ms. The model retains its GT target kinematics and robot-state history. These diagnostic examples characterize offline prediction errors; they do not measure closed-loop contact or task success.

Table S16. Completed training-recipe study, four epochs per run. G uses frozen DINOv3-S and I a trainable CNN; both use the corresponding P07 combination. Batch 64 jointly changes learning rate and device count (details in text). Training time excludes saving and validation. Job time runs from job creation through complete evaluation, including the observed queue and startup.

| Profile | Seed | Batch | GPUs | Success /1,400 | SR (%) | Train (min) | Job (min) |
|---|---|---|---|---|---|---|---|
| G | 0 | 16 | 16 | 849 | 60.64 | 263.40 | 301.28 |
| G | 0 | 64 | 32 | 855 | 61.07 | 99.27 | 121.32 |
| G | 1 | 16 | 16 | 834 | 59.57 | 263.28 | 303.58 |
| G | 1 | 64 | 32 | 822 | 58.71 | 97.48 | 120.23 |
| I | 0 | 16 | 16 | 935 | 66.79 | 220.75 | 257.20 |
| I | 0 | 64 | 32 | 884 | 63.14 | 78.86 | 101.74 |
| I | 1 | 16 | 16 | 857 | 61.21 | 224.91 | 258.42 |
| I | 1 | 64 | 32 | 935 | 66.79 | 79.80 | 101.02 |

Table S17. Training-recipe successes per task (100 scenes per entry). B16 and B64 include their respective learning-rate/device settings; s0/s1 denote training seeds. All runs use the same accepted scene identities and inference seeds. Differences across tasks remain visible despite similar totals.

| | G: frozen DINOv3-S | | | | I: trainable CNN | | | |
|---|---|---|---|---|---|---|---|---|
| Task | B16/s0 | B64/s0 | B16/s1 | B64/s1 | B16/s0 | B64/s0 | B16/s1 | B64/s1 |
| Grasp | 92 | 87 | 91 | 89 | 90 | 95 | 87 | 96 |
| Belt | 100 | 100 | 100 | 97 | 97 | 100 | 99 | 100 |
| Occlusion | 100 | 99 | 100 | 100 | 97 | 98 | 97 | 100 |
| SO(3) | 87 | 88 | 68 | 53 | 86 | 75 | 56 | 55 |
| XYZ | 75 | 58 | 60 | 69 | 82 | 69 | 77 | 85 |
| SE(3) placement | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| Trans. placement | 1 | 10 | 10 | 4 | 4 | 5 | 13 | 29 |
| Full passage | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Phase passage | 0 | 0 | 13 | 0 | 72 | 32 | 14 | 46 |
| Can | 99 | 99 | 97 | 98 | 100 | 98 | 99 | 100 |
| Slider | 83 | 81 | 76 | 80 | 81 | 86 | 75 | 89 |
| Sphere | 40 | 50 | 50 | 57 | 56 | 47 | 60 | 73 |
| Tabletop sphere | 100 | 99 | 99 | 100 | 98 | 99 | 99 | 100 |
| Replan | 72 | 84 | 70 | 75 | 72 | 80 | 80 | 62 |
| Total | 849 | 855 | 834 | 822 | 935 | 884 | 857 | 935 |

normalization, five Euler steps, and synchronous single-step execution remain common.

**Supervision and offline evaluation.** The original 11,200 training and 350 validation demonstrations supply 1,324,374 and 41,533 valid issued-command labels; training and validation source seeds are disjoint within each task. Every valid current command is retained, including windows whose later action slots are invalid. Only capture demonstrations contain hold labels: 2,562 in training (0.1934% of all labels) and 79 in validation. The other thirteen tasks have no valid hold labels; all phase-passage training and validation labels are close. This limits what the decoder can learn about useful holds outside capture.

Both heads use seed 0, twenty epochs, AdamW with learning rate $3\times10^{-4}$ and weight decay $10^{-4}$, batch 4,096, gradient clipping at 1, and ordinary cross-entropy without class reweighting. Feature means and variances use training data only. Each head is selected by the lowest validation mean of the three per-class NLLs, choosing the earliest epoch in a tie. Context selects epoch 15 with NLL 0.01330; raw-only selects epoch 20 with NLL 0.19640.

The context head recalls 79/79 hold labels with precision 79/83, versus 75/79 and 75/85 for raw-only (Table S18). It correctly classifies 137/153 one- or two-tick pulses, compared with 93/153; neither head incorrectly maps these pulses to hold. These counts describe frozen features on demonstration observations, not closed-loop policy performance. All three per-class NLLs improve. The 79 hold labels receive one third of the checkpoint-selection metric and account for 74.5% of its reduction; training cross-entropy

Table S18. Frozen-decoder validation confusion matrices: 41,533 valid issued commands from 350 demonstrations. Rows are ground truth; columns are predicted open/hold/close. These are offline classification counts, not policy successes.

| | Raw only | | | Context | | |
|---|---|---|---|---|---|---|
| GT | Open | Hold | Close | Open | Hold | Close |
| Open | 8003 | 1 | 214 | 8132 | 0 | 86 |
| Hold | 0 | 75 | 4 | 0 | 79 | 0 |
| Close | 135 | 9 | 33092 | 105 | 4 | 33127 |

remains unweighted. The context head still makes 86 open-to-close and 105 close-to-open validation errors, so the mean loss does not constrain the cost of an individual closed-loop decision.

**Predefined regression cases.** The context head is tested on eleven known development cases. It succeeds on E5 capture scenes 0/9/14/80/85, E5 replan scenes 49/88, and E7 XYZ scene 64 and replan scene 7; the E7 cases reuse the E5-trained head without further fitting. It fails on E5 capture scene 52 and replan scene 35. Thus 9/11 pass the regression list, below the predefined all-eleven requirement. The planned 140 new scenes with three decoders (420 rollouts) are not run, and the learned head is not adopted. This case count is not an independent success-rate estimate.

Both public action APIs preserve the original outputs and RNG in observation mode. Twenty-eight sign/dead-zone preflight trials reproduce their original commands and outcomes. Host recomputation checks 608 issued commands: maximum CPU-logit error is $3.82\times10^{-6}$ and argmax commands agree exactly. The other six action dimensions and predicted horizons also pass the recorded comparisons. Complete per-frame policy-input tensors were not saved, so this is not a full input-tensor equality check.

**Physical failure sequence.** Figure S4 shows both failed cases. In each, the first intervention reverses an approximately $-1$ open command into $+1$ close; it is not merely suppression of a near-zero output. Capture changes at 0.650 s, followed by opening divergence at 0.666 s and no bilateral contact. Replan applies the altered command at 1.550 s, after its native one-tick queue; opening diverges at 1.556 s. The learned head fails at 1.324 s (tray contact) and 5.174 s (downstream exit), respectively. Sign succeeds in both; the fixed dead zone succeeds in replan but also fails in capture. Recorded predictions, commands, and 2 ms physical fields match their references before intervention. Subsequent arm and object divergence reflects closed-loop feedback. These records establish failures of this readout under the recorded conditions; they do not test a future-to-action prediction interface or establish decoder performance on unseen scenes.

**Retained-feature diagnostics.** At the first reversal in capture 52 and replan 35, replacing only the action hidden state with the preceding frame's value changes the close-minus-open logit margin from $+2.452/+0.700$ to $-13.839/-9.811$. Conversely, inserting the current hidden state into the preceding feature vector yields $+2.243/+3.184$; replacing only the raw value does not remove either reversal. These substitutions measure a local response of the frozen classifier to mixed features, which need not correspond to realizable observations or successful executions. The first reversal precedes any prior intervention by this head. The retained data do not separate demonstration-to-policy feature mismatch from sampling sensitivity of the final Euler hidden state; both explanations remain untested. The failed regression result is unchanged.

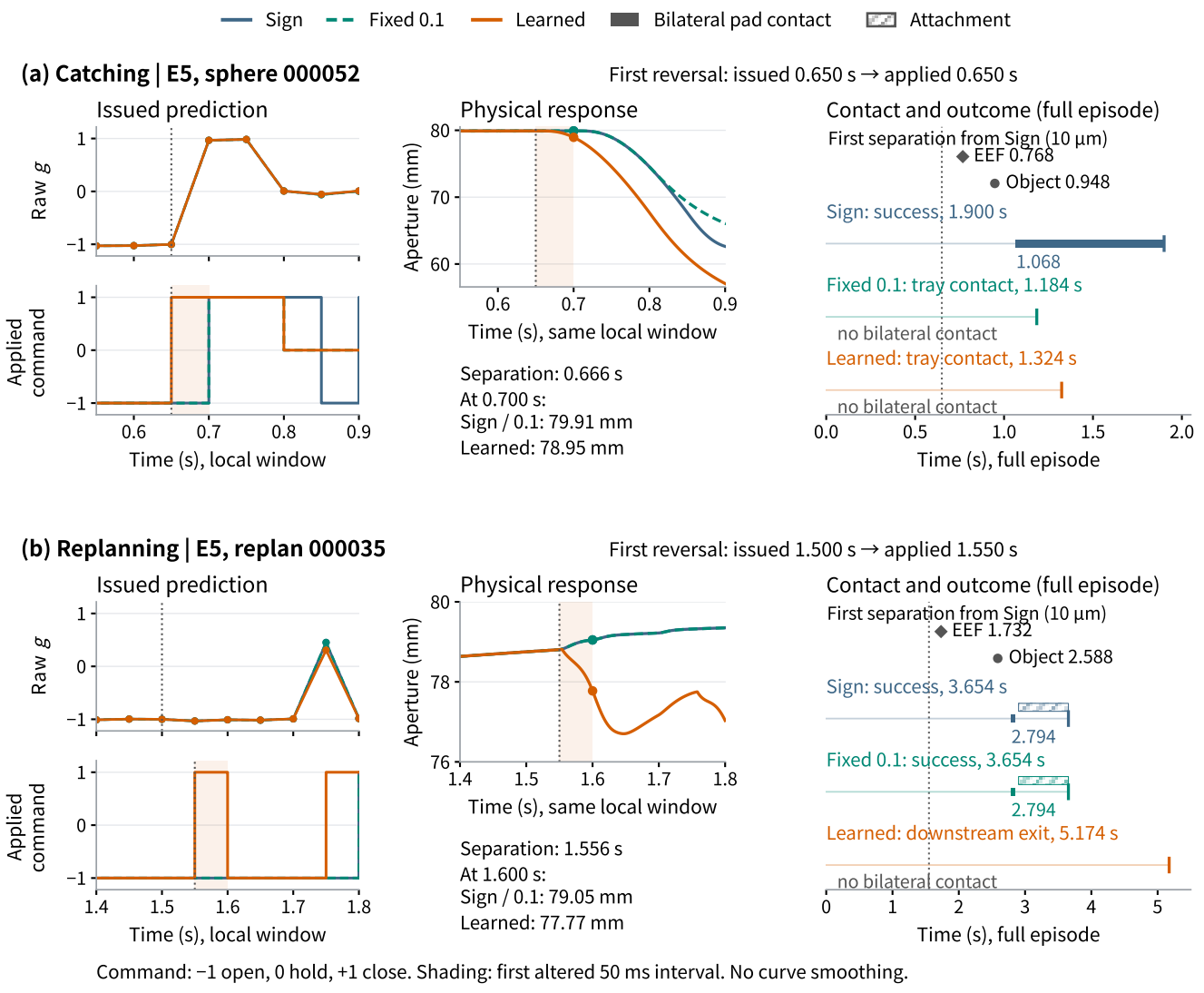



Figure S4. Frozen gripper-readout failures on the two failed predefined regression cases. Commands and measured opening use physical time; event strips retain each run's contact and terminal outcome. Sign, the fixed 0.1 dead zone, and the learned context head use the same frozen policy. The head's first intervention reverses opening into closing, before the observed physical divergence. Detailed windows and full-run events serve different time ranges, labeled on their axes. Contact refers to actual bilateral collision; replan attachment is a separate event. Raw predictions are sampled at 20 Hz and physical states every 2 ms. Separation of the learned run from Sign uses a 10 $\mu$m aperture or componentwise position threshold. The cases are selected development diagnostics, not new-scene success estimates or future-prediction interventions.

## References

[1] Heng Fang, Shangru Li, Shuhan Wang, Xuanyang Xi, Dingkang Liang, and Xiang Bai. Towards generalizable robotic manipulation in dynamic environments. In *European Conference on Computer Vision*, 2026. 18

[2] Gunnar Farnebäck. Two-frame motion estimation based on polynomial expansion. In *Proceedings of the Scandinavian Conference on Image Analysis*, pages 363–370, 2003. 8

[3] Alexander Khazatsky, Karl Pertsch, Suraj Nair, et al. DROID: A large-scale in-the-wild robot manipulation dataset. *arXiv preprint arXiv:2403.12945*, 2024. 17

[4] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. *arXiv preprint arXiv:2306.03310*, 2023. 17, 18

[5] Yushan Liu, Jingjing Fan, Shoujie Li, Yifan Xie, Xiao-Ping Zhang, and Wenbo Ding. PACT-WAM: Predicting Actions and Visual Foresight with Compact Temporal Encoding for Robot Manipulation. *arXiv preprint arXiv:2602.15882*, 2026. 7

[6] Yunfan Lou, Hewen Gao, Xiyu Zhu, Zhuoran Qiao, Xuan Han, Yifan Yang, Yifan Ye, Boxian Yao, and Zhibo Pang. DynamicWAM: Dual-path motion conditioning for world-action models in dynamic manipulation. *arXiv preprint arXiv:2608.00793*, 2026. 6

[7] Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025. 6

[8] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, et al. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025. 5

[9] Yuran Wang, Siqiao Huang, Mingleyang Li, Chenhao Zhang, Jiaqi Liang, Weiyang Jin, Yue Chen, Xuemin Chi, Donghao Zhou, Qize Yu, Yu-Kai Wang, Yuhan Rui, Shenzhe Yao, Zhen Yuan, Zhenhao Shen, Kefei Zhu, Zijie Zhu, Ning Gao, Xiaowei Chi, Guanqi He, Shanghang Zhang, Hao Dong, Lin Shao, and Hang Zhao. OpenWAM: An Open, Modular Exploration Towards Systematic World-Action Model Pretraining. *arXiv preprint arXiv:2609.07398*, 2026. 6

[10] Wenyao Zhang, Hongsi Liu, Zekun Qi, Yunnan Wang, Xin-Qiang Yu, Jiazhao Zhang, Runpei Dong, Jiawei He, He Wang, Zhizheng Zhang, Li Yi, Wenjun Zeng, and Xin Jin. DreamVLA: A vision-language-action model dreamed with comprehensive world knowledge. In *Advances in Neural Information Processing Systems*, pages 24195–24228, 2025. 6